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Wide-Area Illumination Panels Employing Flexible Printed Circuits and Side-emitting LEDs

Abstract

A backlight for an LCD display has an optically transmissive sheet with first and second broad-area surfaces, four edges, a light mixing region adjacent to a first edge, and a light extraction region. The mixing region has smooth planar surfaces for guiding light via total internal reflection. Light extraction features are distributed across the extraction region with increasing density toward the opposing edge. A flexible LED strip extends parallel to the first edge, with a flexible circuit and side-emitting LED packages of less than 2 mm in height. The LED packages have light emitting surfaces perpendicular to the circuit's major surface. A metallic housing partially encases the LED array with a linear channel providing light passage to the mixing region. At least one extraction feature has a surface relief microstructure separated from other features and edges by smooth planar portions of the sheet.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of U.S. patent application Ser. No. 18/513,474, filed Nov. 17, 2023, which is a continuation of U.S. patent application Ser. No. 17/485,498, filed Sep. 27, 2021, which is a continuation of U.S. patent application Ser. No. 17/203,663, filed Mar. 16, 2021, which is a continuation of U.S. patent application Ser. No. 16/828,941, filed Mar. 24, 2020, which is a continuation of U.S. patent application Ser. No. 16/016,595, filed Jun. 23, 2018, which is a continuation of U.S. patent application Ser. No. 13/766,698, filed Feb. 13, 2013, which claims priority from U.S. provisional application Ser. No. 61/598,854 filed on Feb. 14, 2012, the disclosure of which is incorporated herein by reference in its entirety.

NOTICE OF MATERIAL SUBJECT TO COPYRIGHT PROTECTION

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BACKGROUND OF THE INVENTION

1. Field of the Invention

[0003] The present invention relates to planar light emitting waveguides such planar plate or slab waveguides distributing light along the broad surface of the plate and emitting the distributed light from the broad-area plate surface. This invention also relates to an apparatus and method of inputting light into a planar waveguide through its face as opposed to edge-lit light guide panels where light is input through one of the waveguide edges. More particularly, this invention relates to

panel luminaires, illuminated panel signs, illuminated window pane signs, front lights, backlights, lighting panels, LCD display backlights, computer screens, advertising displays, road signs, and the like, as well as to a method for redistributing light from a variety of light sources.

2. Description of Background Art

[0004] Conventionally, light emitting devices employing a planar waveguide include an optically transmissive plate, a light source coupled to the plate's edge and a series of optical features distributed along a major surface of the plate for extracting light at predetermined locations of the surface.

[0005] However, a number of applications exist where edges of the waveguide are not accessible or it is otherwise impractical to input light through an edge. Furthermore, many existing structural or artistic articles which can provide light guiding and distribution functions are not always readily transformable to edge-lit applications. Typical examples include framed glass windows of building facades and doors, storefront window panes, as well as various interior and exterior architectural features employing transparent glass or plastic panels.

[0006] Other examples of common objects which could be used as planar waveguides but may not be suitable for light input from an edge include but are not limited to planar slabs of glass or transparent plastic which edges are roughened or sanded. In a yet further example, the edges of some transparent slabs or panels may be tapered making it difficult to input light from a relatively large source. The light input aperture of edges may also be too small compared to the size of the light source due to the insufficient thickness of the transparent slab or panel.

[0007] It is therefore an object of this invention to provide an improved illumination system providing an efficient light input through a face of a planar waveguide as opposed to light input through an edge. It is another object of this invention to provide a convenient light injection into a planar waveguide, such as an existing window pane of a building, through its face, without having to penetrate into the waveguide's surface. It is yet another object of this invention to provide convenient light input in one area of a major surface of a planar waveguide and light extraction from the waveguide in another area of the surface. It is yet another object of this invention to provide an improved method of coupling light to a planar waveguide without having to access its edges and while substantially reducing or eliminating the unwanted light spillage due to the coupling. Other objects and advantages of this invention will be apparent to those skilled in the art from the following disclosure.

BRIEF SUMMARY OF THE INVENTION

[0008] Accordingly, the present invention is directed to face-lit planar waveguide illumination systems which may be employed to redistribute light emitted by a compact light source over a large area of the planar waveguide and re-emit at least a portion of the distributed light from a major surface of the waveguide. More particularly, this invention is directed to a system for injecting light into the face of a planar plate, slab or substrate of an optically transmissive material in one area and extracting at least some of the injected light from another area of the plate or substrate.

[0009] The present invention solves a number of problems associated with light distribution and illumination using planar waveguides by providing a face-lit solution which is not hindered by the limitations of conventional edge-lit illumination devices requiring the access to waveguide edges or surface penetration for enabling light input.

[0010] An advantage of the present system is to provide controlled light input through a face of the planar waveguide in one location so that such light can be propagated along the waveguide in response to the optical transmission and a total internal reflection generally towards at least one well-defined direction and can then be extracted from another location of the waveguide along said direction.

[0011] Light is injected into the planar waveguide by means of an elongated (linear) optical element attached to a face of the waveguide and optically coupled to said face.

[0012] According to one aspect of the invention, light is input into a face or broad-area surface the

planar waveguide non-invasively and without penetrating into the waveguide's body. According to another aspect of the invention, a linear configuration of the optical element allows for coupling light into the waveguide from a linear light source which may be represented by an elongated light-emitting element or a linear array of compact light-emitting elements. According to yet another aspect, the linear configuration of the light coupling optical element may also be used for injecting light into the waveguide from a discrete light source optically coupled to a terminal end of the optical element.

[0013] In at least one embodiment, the invention features a planar optical waveguide exemplified by a portion of an optically transmissive window pane, a linear light coupling optical element attached to a planar face or major broad-area surface of the pane and a linear light source. The light source includes a strip light emitting diodes (LEDs) incorporated is a linear array. The linear LED array is positioned generally parallel to the longitudinal axis of linear optical element and is configured to illuminate the optical element with a beam of light. The linear optical element is optically coupled to the face of the window pane by means of a planar surface and by providing a good optical contact between the surface and the face of the window pane.

[0014] The optical element is configured to inject light into the medium of the window pane at a sufficiently low out-of-plane angle permitting for the subsequent light propagation within by means of a total internal reflection from the opposing faces of the pane. According to an aspect of the present invention, the optical element injects light into the waveguide mode while directing the light beam towards a well-defined direction along the window pane so that the injected light can be mixed along the propagation path and then extracted at another location of the window pane.

[0015] In at least one embodiment, the invention includes one or more light extracting features positioned along the prevailing path of light propagation in the waveguide. The light extracting features are configured to extract light from the waveguide and direct such light towards a surface perpendicular. According to an aspect of the invention, at least a portion of the extracted light may be directed away from the waveguide and towards a viewer thus providing conspicuous visibility of the light extracting features.

[0016] In at least one embodiment, the invention includes a linear collimating element optically coupled to the light source. Various implementations of the collimating element include linear or axisymmetrical refractive lenses, TIR lenses, reflectors, concentrators, and any combinations thereof.

[0017] Various implementation of the light coupling optical element include forming at least a part of the element from an elongated block of transparent material. In one implementation, the elongated block has a shape of a right-angle prism or wedge. In one implementation, the elongated block has at least one planar surface or at least one curvilinear surface. In one implementation, the elongated block has a triangular transversal cross-section. In one implementation, the elongated block has a trapezoidal transversal cross-section. In further implementations, the optical element has sharply asymmetrical and axisymmetrical configurations in a transversal cross-section. Various faces or surfaces of the optical element may be configured for refracting light and/or for reflecting light by means of a specular reflection or TIR.

[0018] In at least one embodiment the light coupling optical element includes a strip of a film material laminated onto the face of the waveguide. Various implementations of such optical element include prismatic films, holographic films, diffractive films, and any other types of light turning films which may be used for suppressing the refraction at the waveguide surface or for injecting light into waveguide at angles allowing for TIR propagation.

[0019] Various implementations of the face-lit waveguide illumination system include suitable housing components and means for blocking stray light.

[0020] In at least one embodiment of the invention, the width of the linear optical element is defined by the thickness of the planar waveguide. According one specific implementation, the width of the optical contact area of the linear optical element with a major surface of the waveguide

is approximately equal to or less than $2d/\sqrt{n^2-1}$, where d is the thickness of the waveguide and n is the refractive index of the waveguide's medium. According to other specific implementations, the transversal width of the light coupling optical element is less than two times the thickness of the planar waveguide and more preferably, 1.8 times the thickness of a glass window pane.

[0021] In at least one embodiment, the light source is selected from the group of light emitting elements including fluorescent lamps, linear arrays of light emitting diodes, incandescent lamps, cold-cathode or compact fluorescent lamps, halogen, mercury-vapor, sodium-vapor, metal halide, electroluminescent lamps or sources, field emission devices, and lasers.

[0022] In at least one embodiment, the face-lit waveguide illumination system includes the light coupling optical element made from light transmitting material and having the shape of a linear rod or bar. Such rod or bar may have various cross-sections including but not limited to square, rectangular, triangular, pentagonal, hexagonal, octagonal, trapezoidal, circular, half-circular, and oval. The cross-section may also include circular segments or sectors.

[0023] In at least one embodiment, the optical element includes an optically transmissive light turning or light redirecting film. Various implementations of such film include microstructured prismatic films, diffractive films, holographic films, and films with internal light redirecting structures.

[0024] In at least one embodiment, the illumination system includes means for extracting light from the planar waveguide. In one implementation, the light extracting means include light scattering or light diffusing features. Such features may be embedded into the waveguide's body formed in a surface of the waveguide or externally attached to the waveguide surface. In other implementations, the light extracting means include a textured surface, indicia and or an image print.

[0025] In at least one embodiment, the light coupling optical element may be configured with wave guiding properties. In different implementations, it may be formed by a relatively long elongated body of a light transmitting material or include one or more optical fibers.

[0026] In at least one embodiment, the light coupling optical element is attached to the face of the waveguide by means of an intermediate film which may have a substantially larger area than the contact area of the optical element. In at least one embodiment, the optical element is attached either directly to the face of the waveguide or to the intermediate film or plate using an optically clear adhesive or encapsulant.

Description

BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWING(S)

[0027] The invention will be more fully understood by reference to the following drawings which are for illustrative purposes only:

[0028] FIG. 1 is a schematic perspective view of a face-lit waveguide illumination system, according to at least one embodiment of the present invention.

[0029] FIG. 2 is a schematic cross-sectional view and raytracing of a face-lit waveguide illumination system portion, showing a wedge-shaped light coupling optical element, according to at least one embodiment of the present invention.

[0030] FIG. 3 is a schematic cross-sectional view and raytracing of a face-lit waveguide illumination system portion, showing a tapered light coupling optical element, according to at least one embodiment of the present invention.

[0031] FIG. 4 is a schematic cross-sectional view and raytracing of a face-lit waveguide illumination system portion, showing a symmetrical configuration of a light coupling optical element, according to at least one embodiment of the present invention.

[0032] FIG. 5 is a schematic cross-sectional view and raytracing of a face-lit waveguide illumination system portion, showing a prismatic film laminated onto a face of a window pane, according to at least one embodiment of the present invention.

[0033] FIG. 6 is a schematic view of a face-lit waveguide illumination system portion, showing an exemplary arrangement of linear light emitting and light coupling elements, according to at least one embodiment of the present invention.

[0034] FIG. 7 is a schematic view of a face-lit waveguide illumination system portion, showing a layered light coupling element including internal corrugated boundary between two layers, according to at least one embodiment of the present invention.

[0035] FIG. 8 is a schematic view of a face-lit waveguide illumination system portion, showing an optical element having curvilinear reflective surfaces, according to at least one embodiment of the present invention.

[0036] FIG. 9 is a schematic cross-sectional view and raytracing of a face-lit waveguide illumination system, showing a reflective surface extending parallel to a face of a window pane, according to at least one embodiment of the present invention.

[0037] FIG. 10 is a schematic cross-sectional view and raytracing of a face-lit waveguide illumination system, showing a light source optically coupled to an edge of a planar light coupling optical element, according to at least one embodiment of the present invention.

[0038] FIG. 11 is a schematic perspective view and raytracing of a face-lit waveguide illumination system, showing a planar light coupling optical element attached to a face of a window pane, according to at least one embodiment of the present invention.

[0039] FIG. 12 is a schematic perspective view and raytracing of a face-lit waveguide illumination system portion, showing a plurality of optically transmissive rods or bars attached to a face of a planar slab waveguide, according to at least one embodiment of the present invention.

[0040] FIG. 13 is a schematic cross-sectional view of a face-lit waveguide illumination system portion, illustrating light injection into a thin planar waveguide, according to at least one embodiment of the present invention.

[0041] FIG. 14 is a schematic view of a face-lit waveguide illumination system portion, illustrating an exemplary method of attaching a light coupling optical element to a face of a planar waveguide, according to at least one embodiment of the present invention.

[0042] FIG. 15A through FIG. 15F, illustrate various exemplary configurations of a linear optical element, according to at least some embodiments of the present invention.

[0043] FIG. 16 illustrate an exemplary arrangement of optical elements and light sources with respect to a face of a window pane, according to at least one embodiment of the present invention.

[0044] FIG. 17 is a schematic perspective view and raytracing of a face-lit waveguide illumination system portion, showing a plurality of optical fibers attached to a face of a planar slab waveguide, according to at least one embodiment of the present invention.

DETAILED DESCRIPTION OF THE INVENTION

[0045] Referring more specifically to the drawings, for illustrative purposes the present invention is embodied in the system generally shown in the preceding figures. It will be appreciated that the system may vary as to configuration and as to details of the parts without departing from the basic concepts as disclosed herein. Furthermore, elements represented in one embodiment as taught herein are applicable without limitation to other embodiments taught herein, and in combination with those embodiments and what is known in the art.

[0046] A wide range of applications exist for the present invention in relation to the collection and distribution of electromagnetic radiant energy, such as light, in a broad spectrum or any suitable spectral bands or domains. Therefore, for the sake of simplicity of expression, without limiting generality of this invention, the term “light” will be used herein although the general terms “electromagnetic energy”, “electromagnetic radiation”, “radiant energy” or exemplary terms like “visible light”, “infrared light”, or “ultraviolet light” would also be appropriate.

[0047] Furthermore, many applications exist for the present invention in relation to distributing light by means of a planar optical waveguide which hereinafter may also be referenced to as a light guide. The planar optical waveguide refers to a broad class of objects employing an optically transmissive material confined between two opposing broad surfaces which are substantially parallel to each other. The term substantially parallel generally includes cases when the opposing surfaces are parallel within a predetermined accuracy, particularly including the cases when the body of the material defined by the above surfaces has a slightly tapered shape or has a variable thickness across the surface.

[0048] According to the present invention, the planar waveguide may be exemplified by a transparent plate, slab, panel, pane, light-transmitting substrate or any suitable sheetform of an optically transmissive material. This invention is also applicable to any two-dimensional shape variations of the sheetforms, including but not limited to a rectangle, a polygon, a circle, a strip, a freeform, or any combination therein. This invention is further applicable to any three-dimensional shapes that can be obtained by bending the sheetforms accordingly, including but not limited to cylindrical or semi-cylindrical shapes, conical shapes, corrugated shapes, and the like.

[0049] The present invention seeks to provide illumination systems capable of coupling light into a planar waveguide through its broad-area front or back surface in one location and extracting the coupled light at a different location of the waveguide.

[0050] According to the present invention, there is provided an illumination system employing a waveguide exemplified by an optically transmissive panel which may also hereinafter be referred to as a light guide panel or LGP. The panel is made from a material which has a refractive index greater than that of the outside medium and is therefore inherently capable of guiding light within the panel by means of a Total Internal Reflection (TIR) from its opposing broad surfaces, provided that the internal incidence angles onto either of the surfaces are greater than a critical angle of TIR characterizing the surfaces.

[0051] For the purpose of this discussion, the term "incidence angle" of a light ray in relation to a surface generally refers to an angle that this ray makes with respect to a normal to that surface. It will be appreciated by those skilled in the art of optics that, when referring to light or other waves passing through a boundary formed between two different refractive media, such as air and glass, for example, the ratio of the sines of the angles of incidence and of refraction is a constant that depends on the ratio of refractive indices of the media (the Snell's law of refraction). The following relationship can describe a light bending property of an interface between two refractive media: $n_{\text{sub.I}} \sin \phi_{\text{sub.I}} = n_{\text{sub.R}} \sin \phi_{\text{sub.R}}$, where $n_{\text{sub.I}}$ and $n_{\text{sub.R}}$ are the respective refractive indices of the materials forming the optical interface and $\phi_{\text{sub.I}}$ and $\phi_{\text{sub.R}}$ are the angle of incidence and the angle of refraction, respectively. It will be further appreciated that such optical interface can also be characterized by a critical TIR angle which is the value of $\phi_{\text{sub.I}}$ for which $\phi_{\text{sub.R}}$ equals 90° . Accordingly, for a surface characterized by a stepped drop in refractive index along the propagation path of a ray, the incidence angle may be less than, equal to, or greater than the TIR angle at the given surface.

[0052] A TIR angle $\phi_{\text{sub.TIR}}$ can be found from the following expression:

$$\phi_{\text{TIR}} = \arcsin(n_R / n_I \cdot \text{Math. sin}90^\circ) = \arcsin(n_R / n_I) \quad (\text{Equation1})$$

[0053] In an exemplary case of the interface between glass with the reflective index $n_{\text{sub.I}}$ of about 1.51 and air with $n_{\text{sub.R}}$ of about 1, $\phi_{\text{sub.TIR}}$ is approximately equal to 41.5° .

[0054] It will be appreciated that, once light is input into the planar light guide, such as LGP, and its propagation angles permit for TIR to occur at LGP's longitudinal walls, the light becomes trapped in LGP and can propagate considerable distances until it is extracted, absorbed or reached an edge of the panel.

[0055] The present invention will now be described by way of example with reference to the accompanying drawings.

[0056] FIG. 1 depicts an embodiment of a face-lit waveguide illumination system 2 in accordance with the invention. System 2 includes a planar light guide exemplified by a rectangular glass window pane 4 which is mounted within a rigid frame 300. Window pane 4 thus has two opposing parallel broad-area surfaces which are exposed to the air and four edges which are conventionally covered by frame 300.

[0057] While there are many uses for the face-lit illumination system of this invention, the embodiments described here are directed particularly to using various optically transparent panels or panes which are parts of the existing exterior or interior structures and/or decorative elements of buildings. Useful examples of such panels include but are not limited to framed or unframed vertical wall windows, door windows, glass or transparent-plastic facades of buildings, glazed openings in walls and ceilings, vertical or horizontal interior space partitions, screens, and the like. It will be appreciated that such types of panels, albeit often providing a good optical transparency, are not commonly designed for illumination purposes. Yet, many of them represent convenient existing structures which are highly transparent and shaped in the form of a plate having smooth broad-area surfaces, which can make them usable for illumination purposes or for providing various lighting effects to an observer. It is therefore an object of this invention to provide means for utilizing such already existing optically-transmissive structures as planar optical waveguides transporting light from one location to another so that light can be injected into the face of the respective waveguide, distributing such light through the waveguide's body and emit the injected light from another location of the waveguide without having to access its edges for light input.

[0058] Accordingly, by way of example and not limitation, window pane 4 may be of the type typically found in single-pane or dual-pane windows or doors of the residential or commercial buildings. In another non-limiting example, window pane 4 may be a part of a storefront window in a shopping mall. It will be appreciated that such windows or doors, although being typically transparent and optically clear, generally have at least some or all of the edges of the window pane covered with a frame and are therefore not suitable for conventional, edge-lit applications.

[0059] System 2 of FIG. 1 further includes a light coupling optical element 6 made from an optically transmissive material and configured for injecting light into pane 4 through its face as opposed to injecting light through an edge in edge-lit illumination systems. Optical element 6 is attached to one of the two opposing broad-area surfaces or faces of window pane 4 and is optically coupled to the pane with refractive index matching. According to the preferred embodiments, it is important that optical element 6 is attached to the face of window pane 4 externally, that is without surface penetration.

[0060] The term “optically coupled” is directed to mean any relationship between two optical components which enables light to pass from one optical component to the other without being redirected back or absorbed. The term “refractive index matching”, in relationship to an optical coupling, is directed to mean such optical coupling between two refractive optical components where the refractive indices of both components are selected to approximate each other.

[0061] In the context of the present invention, index matching may serve a congruence of goals. One particular goal of the index matched optical coupling is reducing or eliminating the Fresnel reflections at the respective optical interface in a broad range of incidence angles. Another goal may be eliminating or suppressing TIR in an exemplary case where the refractive index of a first optical element is less than that of the second optical element and where the light is designed to propagate from the first element to the second element. In this case, the proper index matching should ensure that no TIR will occur at the respective optical interface within the designed range of incidence angles and that essentially all of the light will be transmitted to the second optical element. It should be understood that index matching generally precludes forming any air gap between the two refractive optical elements.

[0062] For the purpose of this invention and from the practical standpoint, two optical elements may also be considered optically coupled with index matching when they are disposed in a tight

optical contact, directly or through one or more intermediate optical layers, and when the light-receiving optical element has a greater refractive index than the light-emitting component or, if lower, when the differences between the refractive indices are less than a predefined sufficiently small amount.

[0063] According to one embodiment, optical element **6** may be glued to the surface of pane **4** using an optically clear adhesive which refractive index is greater or approximately equal to the refractive index of the material of element **6**. The adhesive may have a relatively high tack for permanent bond or a relatively low tack permitting for the eventual removal of element **6** from pane **4**. According to one embodiment, optical element **6** may be coupled to pane **4** using an intermediate optically clear substrate which has a similar refractive index and which, in turn, is attached to the surface of pane **5** with a good optical contact.

[0064] Referring further to FIG. **1**, optical element **6** has an elongated shape with a longer dimension being substantially greater than the shorter dimensions and is exemplified by a linear prism having a triangular transversal cross-section. Suitable materials for making such linear prism include but are not limited to water-clear glass, Poly(methyl methacrylate) (PMMA), polycarbonate, styrene, cured urethane, silicone, and the like. Optical element **6** is positioned near a bottom edge of window pane **4** with its longitudinal axis **44** aligned generally parallel to the bottom edge of the pane and perpendicular to an intended prevailing direction **98** of light propagation. In the case illustrated in FIG. **1**, direction **98** generally points from the bottom edge of pane **4** towards its top edge.

[0065] It is noted that the illustrated position and orientation of optical element **6** with respect to the edges of pane **4** should not be constructed as limiting this invention in any way. Element **6** may also be positioned at any other location of the optically transmissive surface of pane **4**. Particularly, element **6** may be located at and aligned with respect to any other edge of pane **4**, including the vertical edges and the top edge. In one embodiment, optical element **6** may also be located at a considerable distance from all of the edges, including the case when it is located near the center of pane **4**. Additionally, linear optical element **6** may also be disposed in any other suitable orientation of its longitudinal axis **44** with respect to the edges of pane **4**, including perpendicular and angular positions.

[0066] Referring yet further to FIG. **1**, system **2** additionally includes a light source **30** exemplified by a linear array or a strip of light emitting diodes **32** illuminating at least one longitudinal face of the prism that forms optical element **6**. Optical element **6** is designed to intercept at least a substantial portion of light emitted by source **30** and communicate such light into pane **4** so that light source **30** becomes optically coupled to pane **4** by means of element **6**.

[0067] It should be understood that light source **30** is not limited to light emitting diodes (LEDs) and may include any suitable single or multiple light sources of any known type, including but not limited to: fluorescent lamps, incandescent lamps, cold-cathode or compact fluorescent lamps, halogen, mercury-vapor, sodium-vapor, metal halide, electroluminescent lamps or sources, field emission devices, lasers, etc. In a linear configuration, light source **30** may comprise a single linear light-emitting element or may include two or more compact light emitting elements incorporated into a linear array. When light source **30** includes multiple light emitting elements, each of such elements may have any suitable shape, including compact or extended two-dimensional or one-dimensional (elongated) shapes.

[0068] Light source **30** may further include integrated or external optics such as collimating or light-redistributing lenses, mirrors, lens arrays, mirror arrays, light diffusers, waveguides, optical fibers and the like. When multiple light emitting elements are employed, each of such elements may be provided with individual optics. Alternatively, a single linear optic may be provided for the entire array to collimate light or otherwise shape the emitted beam in a plane which is perpendicular to the longitudinal axis of the array.

[0069] Referring yet further to FIG. **1**, an optional housing **302** may be provided to cover or encase

light source **30** and/or optical element **6**. By way of non-limiting example, housing **302** may have different functions including but not limited to structural, protective (from dust, moisture, elements, impact, etc.) and/or aesthetic.

[0070] Light source **30** may be provided or associated with an electric power source exemplified by an AC/DC power supply **306** connected to source **30** and an AC power grid using an electric cord **304**. The power source may comprise any conventional power sources used to power various light sources. The power sources may include, for example, one or more electric power supplies, batteries, transformers, LED drivers, various other types of voltage and/or current converters or power conditioning units.

[0071] System **2** further includes a light emitting component attached to a face of pane **4** and including one or more light extracting features **20**. Light extracting features **20** may include any light redirecting structure or device configured for extracting light from pane **4**.

[0072] In an embodiment illustrated in FIG. **1**, the light emitting component is exemplified by a rectangular piece of optically transmissive film **8** applied to a surface of pane **4**. Light extracting features **20** are exemplified by light-scattering areas formed in or attached to film **8**. Each of the light-scattering areas has a two-dimensional shape or outline representing an individual letter so that the series of light-scattering areas represent the word "OPEN".

[0073] It should be understood, however, that the light emitting areas are not limited to letters, words or characters and may have any other two-dimensional shapes which, in turn, may be arranged in any suitable manner. By way of non-limiting examples, features **20** may also represent any geometric shapes, symbols, indicia, images or patterns. Alternatively, features **20** may represent no particular shape or pattern and may simply provide a uniform or non-uniform glow for illumination or visual effect function.

[0074] Film **8** may be applied to the same face of pane **4** as the optical element **6** or to the opposing face. Film **8** should be applied to the respective face with a good optical contact which should be sufficient to enable the interaction of light that may propagate in pane **4** with features **20**.

[0075] The attachment of film **8** to the face of pane **4** may involve the process of lamination with or without adhesive. When no adhesive is used, the material of film **8** should preferably have static cling properties and the contacting surface of film **8** should preferably be calendered to prevent unwanted delamination. In an alternative, film **8** may be laminated onto pane **4** using an optically transmissive adhesive layer. Suitable materials for film **8** include but are not limited to clear or translucent vinyl, acetate, PET, and polyethylene. Many common types of decal films used for signage may also be suitable for film **8**.

[0076] Light extracting features **20** may include any suitable means for receiving and redistributing light and should generally be configured to extract light from pane **4**. Features **20** may be configured to extract light by means of scattering, reflection, refraction, deflection, diffraction and may be further configured to do so with changing one or more properties of light. Exemplary properties of light that may be changed by film **8** and/or features **20** include but are not limited to wavelength, polarization, apparent brightness, spectral distribution, angular and/or spatial distribution, dispersion, etc. According to one embodiment, film **8** is optically clear and configured to generally preserve the spectral properties of light propagating in pane **4**. According to one embodiment, film **8** may have specific color-filtering properties thus being able to change the color of light that passes through the film.

[0077] By way of a more specific illustrative example of light-extracting film **8**, features **20** may include light-diffusing surface microstructures formed in the surface of the film. The microstructures may include ordered or random surface relief features formed, for example, by means of etching, embossing, laser ablation, sanding, micromachining, micro-replication and any other method suitable for producing the desired surface texture or relief. In a further non-limiting example, features **20** may be formed by depositing a layer of light scattering material, such as ink or paint onto the surface of film **8** or directly onto the surface of pane **4**.

[0078] In one embodiment, light extracting features **20** may have phosphorescent or fluorescent properties. For example, features **20** may include one or more shapes cut from a sheet of fluorescent material that converts shorter wavelengths of light in the UV or visible spectrum into longer wavelengths in the visible range. Such sheet-form shapes may be laminated onto the surface of pane **4** and configured to scatter light with fluorescent effect when illuminated with a light source. By way of example and not limitation, the fluorescent material may be configured to convert 350 nm-400 nm UV light from a “black light” into visible wavelengths.

[0079] In one embodiment, light extracting features may include an image print which is made on a sheet of paper or printable polymeric material. Such print should preferably be made using inks or paints that become particularly conspicuous when illuminated by a light source.

[0080] Film **8** and particularly features **20** may be located at any suitable area of pane **4** surface. In at least some embodiments, it may be preferred that features **20** are disposed in a generally different location of the surface of pane **4** than optical element **6** and light source **30**. In one embodiment, features **20** may be located in a mid-portion of pane **4** while light source **30** and/or optical element **6** are located at an edge and masked from viewing. In such an arrangement, a distinct visual appearance of glowing features **20** in the middle of the highly transparent glass window pane may be realized with a minimum glare from light source **30**, thus providing a more conspicuous optical effect.

[0081] The mutual disposition of light coupling optical element **6** and light extracting features **20** should be such that at least a portion of light emitted by source and injected into pane **4** can be intercepted by the light extracting features. In one embodiment, It is also preferred that optical element **6** is designed to inject light into pane **4** while directing such light generally towards the location of features **20**. More particularly, in order to maximize light extraction, it may be preferred that direction **98** generally points towards the center of an area formed by light extracting features **20**.

[0082] In operation, light emitting diodes **32** of source **30** emit light towards light coupling optical element **6** which, in turn, injects said light into pane **4** generally at an acute injection angle with respect to the plane of pane **4** and also directs such light towards light extracting features **20**. The injection angle should be sufficiently low to result the subsequent light propagation between the opposing faces of pane **4** in a waveguide mode. In other words, the incidence angle of light rays onto the opposing faces of pane **4** resulting from the light injection should generally be greater than the TIR angle at each of the faces.

[0083] It will be appreciated that, when the TIR conditions are met at least for one of the two faces of pane **4**, they will generally be met for the opposing parallel face as well. Accordingly, light injected into pane **4** may propagate considerable distances along its broad-area surfaces which may involve a number of light bounces from the opposing faces by means of TIR. The effective distance of light propagation along the pane is primarily defined by the degree of optical transmissivity of the pane' material and by the surface smoothness. It will be appreciated that many common objects such as glass panes of windows and door, acrylic sheets and polycarbonate sheets are generally capable to guide light to distances from at least several centimeters with negligible attenuation up to several meters with some more appreciable attenuation.

[0084] At least some types of commercially available water-clear architectural glass used for storefronts have reduced iron content and often have exceptional light transmission properties. Framed window panes made from such types of glass may be particularly suitable for face-lit waveguide illumination systems disclosed in this invention. As light propagates between the opposing broad-area surfaces of pane **4** towards features **20** in response to transmission and TIR, at least a portion of the propagating light will reach features **20**. Each of features **20** scatters the incident light so that at least a substantial portion of the scattered light is extracted from pane **4** towards the viewer. As a result, features **20** will appear brighter than the surrounding background surfaces of film **8** and pane **4**.

[0085] It will be appreciated that the apparent brightness and color of features **20** may be controlled by the brightness and color of source **30** thus providing a convenient means of making visually appealing and conspicuous illuminated signs. It will be appreciated that an embodiment illustrated in FIG. **1** may be conveniently used for making various signage using existing building elements, such as windows or door glass panels, and without undue structural intrusion. Also, it is noted that this invention may be applied to a variety of panel lighting and signage applications where edge-lit LGPs have been traditionally employed. Since the light injection components of system **2** can be attached to either front or back surface of the LGP, no access to the panel's edge is necessary, which may enhance the utility of the device, especially in the applications where the access to the edges is difficult or unwanted.

[0086] The light injecting and light extracting components of system **2** may be made removable thus making the device practical for temporary, low-cost signage. For example, film **8** that includes suitable light extracting features can be laminated onto the respective face of pane **4** using a low-tack adhesive or using static cling. When no longer needed, such film **8** may be peeled off and discarded or stored for future use. Similarly, light coupling optical element **6** may be attached to a face of pane **4** using a low-tack adhesive or pressed against the pane surface to eliminate the air gap. An index matched, optically transmissive filler or encapsulant may be used to assist in gapless optical coupling of element **6** to pane **4**. Suitable encapsulant materials may include silicones, ethylene vinyl acetate (EVA) or any other soft and clear medium. Alternatively, optical element may be permanently bonded to an optically clear film which, in turn, may be laminated onto the respective face of pane **4** using static cling or low-tack adhesive for easy removability.

[0087] FIG. **2** depicts, in a cross-section, a portion of optically transmissive window pane **4** and an exemplary configuration of the linear prism forming element **6**. Window pane **4** has parallel broad-area surfaces **10** and **12** opposing each other and representing the opposing faces of the pane. The linear prism is formed by a wedge-shaped block of a transparent material which has a sharply asymmetric configuration in a transversal cross-section with the taper of the respective wedge generally pointing towards direction **98**. Particularly, the prism has the transversal cross-section of a scalene right-angle triangle which hypotenuse is facing surface **10** of pane **4**. In the context of this invention, the term "transversal cross-section" of an elongated or linear body is directed to mean a cross-section in a plane which is perpendicular to the prevailing longitudinal axis of said body.

[0088] By way of example and not limitation, such linear prism may be made by extrusion or injection molding from bulk acrylic (PMMA) or polycarbonate material. In a further non-limiting example, the prism may be made by machining of a rectangular block, plate or slab of acrylic, polycarbonate or glass with the subsequent surface polishing to a high gloss.

[0089] The linear prism which forms optical element **6** has a light input face **14** represented by a shorter cathetus of the respective right-angle triangle, a smooth and planar coupling face **16** represented by the triangle's hypotenuse, and a reflective face **18** represented by the longer cathetus. Faces **14** and **16** and faces **16** and **18** make acute dihedral angles with respect to each other pairwise. Face **16** is attached to surface **10** with a good physical and optical contact so that element **6** becomes optically coupled to pane **4**. A layer **28** of index matching clear adhesive is provided to promote the adhesion and optical contact. The refractive index of layer **28** is preferably selected to be greater or approximately match that of element **6** and should ensure that light can freely pass from element **6** to pane **4** in a broad range of incidence angles, without the risk of TIR at the respective optical interface.

[0090] LED-based light source **30** is provided in the immediate vicinity of face **14** of element **6**. Each of light emitting diodes **32** of light source **30** may have an integrated collimating lens **38** of a refractive type. Lens **38** may have a spherical aperture and may be designed to improve light extraction from LED **32** and/or narrow the angular distribution of light emitted by the respective LED chip. The light output surface of source **30** is disposed adjacent to face **14** so that the emitted light can freely enter into element **6** without substantial light spillage. It will be appreciated that,

when face **14** is planar and is exposed to a non-parallel beam of incident light, said light will undergo refraction so that the angular distribution of the beam will be generally narrower within element **6** than in the air.

[0091] The slope of face **14** with respect to the surface of pane **4** is selected so that at least a substantial portion of light entering element **6** makes sufficiently high angles with respect to a normal to the surface of pane **4**. Particularly, the resulting angles of light propagation should allow for TIR at least from the portions of surfaces **10** and **12** exposed to the low-n outside medium, such as air. Accordingly, light entering pane **4** at such sufficiently high incidence angles will be reflected at least once from surface **12** by means of TIR.

[0092] It will be appreciated that, once the condition for TIR at surface **12** is met, it will also be automatically met for the opposing surface **12** due to the parallelism of surfaces **10** and **12**. Therefore, any ray propagating at sufficiently high incidence angle with respect to a surface will become trapped in pane **4** and can propagate in a waveguide mode by bouncing from surfaces **10** and **12**.

[0093] Optical element **6** should preferably be designed to prevent or at least minimize secondary interactions of light rays with face **16** and thus prevent or minimize light decoupling from pane **4** through element **6**. Particularly, rays injected into pane **4** at near-TIR angles should generally strike only a free portion of surface **10** in order to prevent re-entering into element **6** and escaping from the waveguide mode.

[0094] For this purpose, the transversal width of coupling face **16** (represented by the hypotenuse of the respective right-angle triangle in FIG. 2) may be selected to not exceed a certain maximum width $w_{\text{sub.max}}$. The maximum width $w_{\text{sub.max}}$ may be defined from various considerations which may account, for example, for the angular distribution of the beam emitted by source **30**, the transversal size of source **30**, the refractive indices of element **16** and pane **4**, as well as the thickness of pane **4**.

[0095] In one embodiment, the maximum width $w_{\text{sub.max}}$ may be defined from the following relationship:

$$[00002] \quad w_{\text{max}} = \frac{d}{2 \tan(\phi_{\text{TIR}})}, \quad (\text{Equation 2})$$

where d is the thickness of pane **4** and $\phi_{\text{sub.TIR}}$ is the critical TIR angle characterizing surfaces **10** and **12** of the pane. Considering that the medium surrounding pane **4** can commonly be air ($n_{\text{sub.R}} \sim 1$) and that Equation 1 then translates into $\sin(\phi_{\text{sub.TIR}}) = 1/n_{\text{sub.I}}$, obtain after simplification:

$$[00003] \quad w_{\text{max}} = \frac{2d}{\sqrt{n_{\text{I}}^2 - 1}}. \quad (\text{Equation 3})$$

[0096] In an exemplary of a glass window pane **4**, $n_{\text{sub.I}} \sim 1.51$ and $\phi_{\text{sub.TIR}} \sim 41.5^\circ$, which gives $w_{\text{sub.max}} \sim 1.8d$. To put it differently, the ratio between the transversal width of the contact area of linear optical element **6** with the face of a glass window pane and the thickness of such pane should generally not exceed 1.8. At this ratio, the acceptance angle of the optical pair formed by glass pane **4** and linear optical element **6** may be maximized since any ray which enters into pane **4** at greater-than-TIR incidence angle will be captured into the waveguide mode. In more general terms, the width of the elongated body of optical element **6** should normally be less than twice the thickness of pane **4** it is attached to.

[0097] In a further, more specific example of a 6-mm window pane, the transversal width of face **16** should generally not exceed 10-11 mm. At the same time, it is noted that the transversal width of input surface **14** (represented by the length of the shorter cathetus of the respective triangle in FIG. 2) should be made sufficient to accommodate the light emitting aperture of light source **30**.

[0098] It will be appreciated from the above examples that a thicker window pane **4** will generally accommodate a wider contact surface of linear light-injecting optical element **6** and a larger-aperture source **30**. However, it should be understood that other factors may also influence the size

selection for element **16** and/or light source **30**. Particularly, when the light beam from source **30** can be collimated to a narrow angular spread, the angular distribution of light in the waveguiding plate of pane **4** may also be made sufficiently narrow. A sufficiently narrow beam, in turn, may be coupled into pane **4** at lower out-of-plane angles thus extending the distance between the individual bounces of a light ray from surface **10** and generally allowing for the use of a wider optical element **6**.

[0099] Referring further to FIG. **2**, surface **18** is configured to intercept rays propagating away from surface **16** and redirect them into pane **4**. The angular disposition of face **18** with respect to faces **14** and **16** is so selected that that at least a substantial portion of light striking surface **18** is reflected by means of TIR with minimum or no light exiting into the surrounding medium. Additionally, the angle of face **18** should be selected so that the reflected rays obtain sufficiently high angles with respect to a normal to surfaces **10** and **12**, generally above the respective TIR angle, and thus can also be coupled into the waveguide mode.

[0100] It will be appreciated that light coupled to pane **4** according to the above principles will propagate along the opposing pane faces away from light source **30** and along direction **98** by bouncing from opposing smooth broad surfaces **10** and **12**. At least a portion of such light may eventually reach light extracting features **20** (not shown in FIG. **2**) located at a distance from source **30** along direction **98**. Light extracting features **20** may thus extract (decouple) light from the pane thus providing the desired illumination function.

[0101] FIG. **1** is a schematic perspective view of a face-lit waveguide illumination system, according to at least one embodiment of the present invention.

[0102] FIG. **2** is a schematic cross-sectional view and raytracing of a face-lit waveguide illumination system portion, showing a wedge-shaped light coupling optical element, according to at least one embodiment of the present invention.

[0103] FIG. **3** is a schematic cross-sectional view and raytracing of a face-lit waveguide illumination system portion, showing a tapered light coupling optical element, according to at least one embodiment of the present invention.

[0104] FIG. **4** is a schematic cross-sectional view and raytracing of a face-lit waveguide illumination system portion, showing a symmetrical configuration of a light coupling optical element, according to at least one embodiment of the present invention.

[0105] FIG. **5** is a schematic cross-sectional view and raytracing of a face-lit waveguide illumination system portion, showing a prismatic film laminated onto a face of a window pane, according to at least one embodiment of the present invention.

[0106] FIG. **6** is a schematic view of a face-lit waveguide illumination system portion, showing an exemplary arrangement of linear light emitting and light coupling elements, according to at least one embodiment of the present invention.

[0107] FIG. **7** is a schematic view of a face-lit waveguide illumination system portion, showing a layered light coupling element including internal corrugated boundary between two layers, according to at least one embodiment of the present invention.

[0108] FIG. **8** is a schematic view of a face-lit waveguide illumination system portion, showing an optical element having curvilinear reflective surfaces, according to at least one embodiment of the present invention.

[0109] FIG. **9** is a schematic cross-sectional view and raytracing of a face-lit waveguide illumination system, showing a reflective surface extending parallel to a face of a window pane, according to at least one embodiment of the present invention.

[0110] FIG. **10** is a schematic cross-sectional view and raytracing of a face-lit waveguide illumination system, showing a light source optically coupled to an edge of a planar light coupling optical element, according to at least one embodiment of the present invention.

[0111] FIG. **11** is a schematic perspective view and raytracing of a face-lit waveguide illumination system, showing a planar light coupling optical element attached to a face of a window pane,

according to at least one embodiment of the present invention.

[0112] FIG. **12** is a schematic perspective view and raytracing of a face-lit waveguide illumination system portion, showing a plurality of optically transmissive rods or bars attached to a face of a planar slab waveguide, according to at least one embodiment of the present invention.

[0113] FIG. **13** is a schematic cross-sectional view of a face-lit waveguide illumination system portion, illustrating light injection into a thin planar waveguide, according to at least one embodiment of the present invention.

[0114] FIG. **14** is a schematic view of a face-lit waveguide illumination system portion, illustrating an exemplary method of attaching a light coupling optical element to a face of a planar waveguide, according to at least one embodiment of the present invention.

[0115] FIG. **15A** through FIG. **15F**, illustrate various exemplary configurations of a linear optical element, according to at least some embodiments of the present invention.

[0116] FIG. **16** illustrate an exemplary arrangement of optical elements and light sources with respect to a face of a window pane, according to at least one embodiment of the present invention.

[0117] FIG. **17** is a schematic perspective view and raytracing of a face-lit waveguide illumination system portion, showing a plurality of optical fibers attached to a face of a planar slab waveguide, according to at least one embodiment of the present invention. illustrates an alternative configuration of optical element **6** which comprises planar faces **14**, **16**, **18** and further comprises a face **22** which extends generally parallel to face **16** and defines a planar portion of the optical element. The planar portion of element **6** is terminated with a wedge portion defined by face **18**.

Face **14** of FIG. **1** is a schematic perspective view of a face-lit waveguide illumination system, according to at least one embodiment of the present invention.

[0118] FIG. **2** is a schematic cross-sectional view and raytracing of a face-lit waveguide illumination system portion, showing a wedge-shaped light coupling optical element, according to at least one embodiment of the present invention.

[0119] FIG. **3** is a schematic cross-sectional view and raytracing of a face-lit waveguide illumination system portion, showing a tapered light coupling optical element, according to at least one embodiment of the present invention.

[0120] FIG. **4** is a schematic cross-sectional view and raytracing of a face-lit waveguide illumination system portion, showing a symmetrical configuration of a light coupling optical element, according to at least one embodiment of the present invention.

[0121] FIG. **5** is a schematic cross-sectional view and raytracing of a face-lit waveguide illumination system portion, showing a prismatic film laminated onto a face of a window pane, according to at least one embodiment of the present invention.

[0122] FIG. **6** is a schematic view of a face-lit waveguide illumination system portion, showing an exemplary arrangement of linear light emitting and light coupling elements, according to at least one embodiment of the present invention.

[0123] FIG. **7** is a schematic view of a face-lit waveguide illumination system portion, showing a layered light coupling element including internal corrugated boundary between two layers, according to at least one embodiment of the present invention.

[0124] FIG. **8** is a schematic view of a face-lit waveguide illumination system portion, showing an optical element having curvilinear reflective surfaces, according to at least one embodiment of the present invention.

[0125] FIG. **9** is a schematic cross-sectional view and raytracing of a face-lit waveguide illumination system, showing a reflective surface extending parallel to a face of a window pane, according to at least one embodiment of the present invention.

[0126] FIG. **10** is a schematic cross-sectional view and raytracing of a face-lit waveguide illumination system, showing a light source optically coupled to an edge of a planar light coupling optical element, according to at least one embodiment of the present invention.

[0127] FIG. **11** is a schematic perspective view and raytracing of a face-lit waveguide illumination

system, showing a planar light coupling optical element attached to a face of a window pane, according to at least one embodiment of the present invention.

[0128] FIG. **12** is a schematic perspective view and raytracing of a face-lit waveguide illumination system portion, showing a plurality of optically transmissive rods or bars attached to a face of a planar slab waveguide, according to at least one embodiment of the present invention.

[0129] FIG. **13** is a schematic cross-sectional view of a face-lit waveguide illumination system portion, illustrating light injection into a thin planar waveguide, according to at least one embodiment of the present invention.

[0130] FIG. **14** is a schematic view of a face-lit waveguide illumination system portion, illustrating an exemplary method of attaching a light coupling optical element to a face of a planar waveguide, according to at least one embodiment of the present invention.

[0131] FIG. **15A** through FIG. **15F**, illustrate various exemplary configurations of a linear optical element, according to at least some embodiments of the present invention.

[0132] FIG. **16** illustrate an exemplary arrangement of optical elements and light sources with respect to a face of a window pane, according to at least one embodiment of the present invention.

[0133] FIG. **17** is a schematic perspective view and raytracing of a face-lit waveguide illumination system portion, showing a plurality of optical fibers attached to a face of a planar slab waveguide, according to at least one embodiment of the present invention. is disposed perpendicular to surface **10** and is configured to accept light in a broad range of incidence angles. Light source **30** comprising a linear array of light emitting diodes **32** is positioned adjacent to face **14**. The optical axis of each light emitting diode **32** is aligned parallel to the plane of pane **4**. Layer **28** of index-matched optical adhesive of any other type of coupling agent is provided along the extent of surface **16** so that there is a good optical contact between element **6** and pane **4**.

[0134] In operation, referring to FIG. **1** is a schematic perspective view of a face-lit waveguide illumination system, according to at least one embodiment of the present invention.

[0135] FIG. **2** is a schematic cross-sectional view and raytracing of a face-lit waveguide illumination system portion, showing a wedge-shaped light coupling optical element, according to at least one embodiment of the present invention.

[0136] FIG. **3** is a schematic cross-sectional view and raytracing of a face-lit waveguide illumination system portion, showing a tapered light coupling optical element, according to at least one embodiment of the present invention.

[0137] FIG. **4** is a schematic cross-sectional view and raytracing of a face-lit waveguide illumination system portion, showing a symmetrical configuration of a light coupling optical element, according to at least one embodiment of the present invention.

[0138] FIG. **5** is a schematic cross-sectional view and raytracing of a face-lit waveguide illumination system portion, showing a prismatic film laminated onto a face of a window pane, according to at least one embodiment of the present invention.

[0139] FIG. **6** is a schematic view of a face-lit waveguide illumination system portion, showing an exemplary arrangement of linear light emitting and light coupling elements, according to at least one embodiment of the present invention.

[0140] FIG. **7** is a schematic view of a face-lit waveguide illumination system portion, showing a layered light coupling element including internal corrugated boundary between two layers, according to at least one embodiment of the present invention.

[0141] FIG. **8** is a schematic view of a face-lit waveguide illumination system portion, showing an optical element having curvilinear reflective surfaces, according to at least one embodiment of the present invention.

[0142] FIG. **9** is a schematic cross-sectional view and raytracing of a face-lit waveguide illumination system, showing a reflective surface extending parallel to a face of a window pane, according to at least one embodiment of the present invention.

[0143] FIG. **10** is a schematic cross-sectional view and raytracing of a face-lit waveguide

illumination system, showing a light source optically coupled to an edge of a planar light coupling optical element, according to at least one embodiment of the present invention.

[0144] FIG. **11** is a schematic perspective view and raytracing of a face-lit waveguide illumination system, showing a planar light coupling optical element attached to a face of a window pane, according to at least one embodiment of the present invention.

[0145] FIG. **12** is a schematic perspective view and raytracing of a face-lit waveguide illumination system portion, showing a plurality of optically transmissive rods or bars attached to a face of a planar slab waveguide, according to at least one embodiment of the present invention.

[0146] FIG. **13** is a schematic cross-sectional view of a face-lit waveguide illumination system portion, illustrating light injection into a thin planar waveguide, according to at least one embodiment of the present invention.

[0147] FIG. **14** is a schematic view of a face-lit waveguide illumination system portion, illustrating an exemplary method of attaching a light coupling optical element to a face of a planar waveguide, according to at least one embodiment of the present invention.

[0148] FIG. **15A** through FIG. **15F**, illustrate various exemplary configurations of a linear optical element, according to at least some embodiments of the present invention.

[0149] FIG. **16** illustrate an exemplary arrangement of optical elements and light sources with respect to a face of a window pane, according to at least one embodiment of the present invention.

[0150] FIG. **17** is a schematic perspective view and raytracing of a face-lit waveguide illumination system portion, showing a plurality of optical fibers attached to a face of a planar slab waveguide, according to at least one embodiment of the present invention, light emitting diode **32** illuminates face **14** by a divergent beam of light. Face **14** intercepts at least a substantial portion of the emitted light and admits said light into optical element **6** with at least some refraction which somewhat narrows the angular beam spread. About half of light entering element **6** is communicated to pane **4** directly through the optical interface defined by face **16**. The rest of light enters pane **4** after first reflecting from face **22** and/or face **18**. It will be appreciated that face **22** extending parallel to surface **10** does not change the propagation angle with respect to a normal to surface **10**. Therefore, rays initially propagating in element **6** at greater than TIR incidence angles will maintain such angles after bouncing from face **22**. The slope of face **18** with respect to surface **10** may be selected so that most rays striking said surface are redirected into pane **4** at angles which permit for TIR propagation and so that light escape is minimized.

[0151] Thus, the configuration of optical coupling depicted in FIG. **1** is a schematic perspective view of a face-lit waveguide illumination system, according to at least one embodiment of the present invention.

[0152] FIG. **2** is a schematic cross-sectional view and raytracing of a face-lit waveguide illumination system portion, showing a wedge-shaped light coupling optical element, according to at least one embodiment of the present invention.

[0153] FIG. **3** is a schematic cross-sectional view and raytracing of a face-lit waveguide illumination system portion, showing a tapered light coupling optical element, according to at least one embodiment of the present invention.

[0154] FIG. **4** is a schematic cross-sectional view and raytracing of a face-lit waveguide illumination system portion, showing a symmetrical configuration of a light coupling optical element, according to at least one embodiment of the present invention.

[0155] FIG. **5** is a schematic cross-sectional view and raytracing of a face-lit waveguide illumination system portion, showing a prismatic film laminated onto a face of a window pane, according to at least one embodiment of the present invention.

[0156] FIG. **6** is a schematic view of a face-lit waveguide illumination system portion, showing an exemplary arrangement of linear light emitting and light coupling elements, according to at least one embodiment of the present invention.

[0157] FIG. **7** is a schematic view of a face-lit waveguide illumination system portion, showing a

layered light coupling element including internal corrugated boundary between two layers, according to at least one embodiment of the present invention.

[0158] FIG. **8** is a schematic view of a face-lit waveguide illumination system portion, showing an optical element having curvilinear reflective surfaces, according to at least one embodiment of the present invention.

[0159] FIG. **9** is a schematic cross-sectional view and raytracing of a face-lit waveguide illumination system, showing a reflective surface extending parallel to a face of a window pane, according to at least one embodiment of the present invention.

[0160] FIG. **10** is a schematic cross-sectional view and raytracing of a face-lit waveguide illumination system, showing a light source optically coupled to an edge of a planar light coupling optical element, according to at least one embodiment of the present invention.

[0161] FIG. **11** is a schematic perspective view and raytracing of a face-lit waveguide illumination system, showing a planar light coupling optical element attached to a face of a window pane, according to at least one embodiment of the present invention.

[0162] FIG. **12** is a schematic perspective view and raytracing of a face-lit waveguide illumination system portion, showing a plurality of optically transmissive rods or bars attached to a face of a planar slab waveguide, according to at least one embodiment of the present invention.

[0163] FIG. **13** is a schematic cross-sectional view of a face-lit waveguide illumination system portion, illustrating light injection into a thin planar waveguide, according to at least one embodiment of the present invention.

[0164] FIG. **14** is a schematic view of a face-lit waveguide illumination system portion, illustrating an exemplary method of attaching a light coupling optical element to a face of a planar waveguide, according to at least one embodiment of the present invention.

[0165] FIG. **15A** through FIG. **15F**, illustrate various exemplary configurations of a linear optical element, according to at least some embodiments of the present invention.

[0166] FIG. **16** illustrate an exemplary arrangement of optical elements and light sources with respect to a face of a window pane, according to at least one embodiment of the present invention.

[0167] FIG. **17** is a schematic perspective view and raytracing of a face-lit waveguide illumination system portion, showing a plurality of optical fibers attached to a face of a planar slab waveguide, according to at least one embodiment of the present invention. also allows for injecting at least a substantial portion of light emitted by source **30** into pane **4** through its planar face. Notably, light becomes trapped within pane **4** by means of TIR so that it can be transported along direction **98** and further extracted by light extracting features **20** (not shown in FIG. **1** is a schematic perspective view of a face-lit waveguide illumination system, according to at least one embodiment of the present invention.

[0168] FIG. **2** is a schematic cross-sectional view and raytracing of a face-lit waveguide illumination system portion, showing a wedge-shaped light coupling optical element, according to at least one embodiment of the present invention.

[0169] FIG. **3** is a schematic cross-sectional view and raytracing of a face-lit waveguide illumination system portion, showing a tapered light coupling optical element, according to at least one embodiment of the present invention.

[0170] FIG. **4** is a schematic cross-sectional view and raytracing of a face-lit waveguide illumination system portion, showing a symmetrical configuration of a light coupling optical element, according to at least one embodiment of the present invention.

[0171] FIG. **5** is a schematic cross-sectional view and raytracing of a face-lit waveguide illumination system portion, showing a prismatic film laminated onto a face of a window pane, according to at least one embodiment of the present invention.

[0172] FIG. **6** is a schematic view of a face-lit waveguide illumination system portion, showing an exemplary arrangement of linear light emitting and light coupling elements, according to at least one embodiment of the present invention.

[0173] FIG. **7** is a schematic view of a face-lit waveguide illumination system portion, showing a layered light coupling element including internal corrugated boundary between two layers, according to at least one embodiment of the present invention.

[0174] FIG. **8** is a schematic view of a face-lit waveguide illumination system portion, showing an optical element having curvilinear reflective surfaces, according to at least one embodiment of the present invention.

[0175] FIG. **9** is a schematic cross-sectional view and raytracing of a face-lit waveguide illumination system, showing a reflective surface extending parallel to a face of a window pane, according to at least one embodiment of the present invention.

[0176] FIG. **10** is a schematic cross-sectional view and raytracing of a face-lit waveguide illumination system, showing a light source optically coupled to an edge of a planar light coupling optical element, according to at least one embodiment of the present invention.

[0177] FIG. **11** is a schematic perspective view and raytracing of a face-lit waveguide illumination system, showing a planar light coupling optical element attached to a face of a window pane, according to at least one embodiment of the present invention.

[0178] FIG. **12** is a schematic perspective view and raytracing of a face-lit waveguide illumination system portion, showing a plurality of optically transmissive rods or bars attached to a face of a planar slab waveguide, according to at least one embodiment of the present invention.

[0179] FIG. **13** is a schematic cross-sectional view of a face-lit waveguide illumination system portion, illustrating light injection into a thin planar waveguide, according to at least one embodiment of the present invention.

[0180] FIG. **14** is a schematic view of a face-lit waveguide illumination system portion, illustrating an exemplary method of attaching a light coupling optical element to a face of a planar waveguide, according to at least one embodiment of the present invention.

[0181] FIG. **15A** through FIG. **15F**, illustrate various exemplary configurations of a linear optical element, according to at least some embodiments of the present invention.

[0182] FIG. **16** illustrate an exemplary arrangement of optical elements and light sources with respect to a face of a window pane, according to at least one embodiment of the present invention.

[0183] FIG. **17** is a schematic perspective view and raytracing of a face-lit waveguide illumination system portion, showing a plurality of optical fibers attached to a face of a planar slab waveguide, according to at least one embodiment of the present invention.) at a different location of the pane.

[0184] It will be appreciated that, besides providing light distribution and transport to features **2**, pane **4** may provide convenient means for mixing and homogenizing light emitted by the LED array. Each of the light emitting diodes **32** may be configured to emit light in the same color in which case features **20** can be uniformly illuminated by that color.

[0185] The LED array may be controlled in a number of ways. In one embodiment, the LED array may be powered on and off by applying the same constant or variable current to each LED.

Alternatively, different LEDs **32** in the array may be configured to emit different colors and each of such different-color LEDs **32** may be controlled individually. Since light beams from individual LEDs **32** mix in pane **4**, this can create various color and brightness effects as light is scattered by the medium of pane **4** or extracted by light extracting features **20**. LEDs **32** may also be arranged in clusters of different-color emitters, such as red-green-blue (RGB) or the like, and the color of the emission may be controlled by changing the intensity of individual LEDs in each cluster.

[0186] FIG. **4** depicts a further alternative configuration of light coupling components of system **2**. Referring to FIG. **4**, there is provided an optically transmissive planar light guide represented by a portion of window pane **4** having opposing broad-area surfaces **10** and **12** extending generally parallel to each other. System **2** further comprises optical element **6** including two identical linear prisms **56** and **356** disposed in a symmetrical configuration. Prisms **56** and **356** have substantially planar faces **216** and **416**, respectively, which are made attachable to surface **10** of window pane **4** with a good optical contact. An optical adhesive or encapsulant may be used to promote such

optical contact.

[0187] A face **214** of prism **56** and a face **314** of prism **356** both are perpendicular to the plane of pane **4**. A face **218** of prism **56** and a face **418** of prism **356** are both sloped with respect to surface **10** so that they can redirect light into pane **4** by means of TIR. A central reflective element is provided between prisms **56** and **356**. The reflective element is formed by symmetrically disposed, sloped reflectors **24** and **29** which are mirrored for good specular reflectivity.

[0188] Light source **30** represented by one or more light emitting diodes **32** is disposed immediately adjacent to the opening between prisms **56** and **356**. An optional heat sink **36** is provided on the back of each light emitting diode **32** to promote natural cooling. Prisms **56** and **356** as well as reflectors **24** and **29** are configured work cooperatively to inject at least a substantial portion of light emitted by LEDs **32** into pane **4** at angles allowing for TIR from surfaces **10** and **12** and towards the respective directions. Accordingly, prism **56** and reflector **24** should be configured to direct light towards direction **98** and prism **356** and reflector **29** should be configured to direct light towards an opposing direction **298**.

[0189] At the same time, the respective prisms and reflectors should be configured to minimize or prevent light re-entry onto light coupling optical element **6**. Accordingly, light emitted by source **230** and injected into pane **4** by element **56** will propagate in pane **4** by means of TIR symmetrically away from the source. Light extracting features **20** (not shown) may be distributed along directions **98** and **298** at any predetermined locations of pane **4**. Such light emitting areas should be configured to extract the injected light towards a viewer or any predetermined direction.

[0190] The transversal width of the area of optical contact of such symmetrical optical element **6** with pane **4** may be defined based on reasoning similar to that used for obtaining Equation 3 above for an asymmetrical linear optical element **6**. Accordingly, in one embodiment, the maximum transversal aperture $A_{\text{sub.max}}$ of the symmetrical element **6** may be selected based on the following relationship:

[00004]
$$A_{\text{max}} = \frac{4d}{\sqrt{n_l^2 - 1}} \cdot \quad (\text{Equation4})$$

[0191] Thus, in an exemplary case of a glass window pane **4** ($n_{\text{sub.I}} \sim 1.51$), the width of the symmetrical light coupling element **6** may be up to a factor of 3.5 greater than the thickness of the glass pane when the acceptance angle is to be maximized.

[0192] Referring to FIG. **4**, it is noted that the illustrated cross-section may also represent various non-linear configurations of optical element **6** and may be applied to the case when optical element **6** has an axisymmetrical configuration which can be obtained by rotating the cross-section of FIG. **4** around an axis.

[0193] The faces of prisms **56** and **356** and the walls of reflectors **24** and **29** are not limited to having a linear profile in a cross-section and may have various concave or convex profiles. Particularly, reflectors **24** and **29** may form a cusped shape which may be well suited for enhancing the efficiency of light coupling into pane **4** or providing a specific light distribution pattern. Likewise, face **18** of FIG. **2** and FIG. **1** is a schematic perspective view of a face-lit waveguide illumination system, according to at least one embodiment of the present invention.

[0194] FIG. **2** is a schematic cross-sectional view and raytracing of a face-lit waveguide illumination system portion, showing a wedge-shaped light coupling optical element, according to at least one embodiment of the present invention.

[0195] FIG. **3** is a schematic cross-sectional view and raytracing of a face-lit waveguide illumination system portion, showing a tapered light coupling optical element, according to at least one embodiment of the present invention.

[0196] FIG. **4** is a schematic cross-sectional view and raytracing of a face-lit waveguide illumination system portion, showing a symmetrical configuration of a light coupling optical element, according to at least one embodiment of the present invention.

[0197] FIG. **5** is a schematic cross-sectional view and raytracing of a face-lit waveguide

illumination system portion, showing a prismatic film laminated onto a face of a window pane, according to at least one embodiment of the present invention.

[0198] FIG. **6** is a schematic view of a face-lit waveguide illumination system portion, showing an exemplary arrangement of linear light emitting and light coupling elements, according to at least one embodiment of the present invention.

[0199] FIG. **7** is a schematic view of a face-lit waveguide illumination system portion, showing a layered light coupling element including internal corrugated boundary between two layers, according to at least one embodiment of the present invention.

[0200] FIG. **8** is a schematic view of a face-lit waveguide illumination system portion, showing an optical element having curvilinear reflective surfaces, according to at least one embodiment of the present invention.

[0201] FIG. **9** is a schematic cross-sectional view and raytracing of a face-lit waveguide illumination system, showing a reflective surface extending parallel to a face of a window pane, according to at least one embodiment of the present invention.

[0202] FIG. **10** is a schematic cross-sectional view and raytracing of a face-lit waveguide illumination system, showing a light source optically coupled to an edge of a planar light coupling optical element, according to at least one embodiment of the present invention.

[0203] FIG. **11** is a schematic perspective view and raytracing of a face-lit waveguide illumination system, showing a planar light coupling optical element attached to a face of a window pane, according to at least one embodiment of the present invention.

[0204] FIG. **12** is a schematic perspective view and raytracing of a face-lit waveguide illumination system portion, showing a plurality of optically transmissive rods or bars attached to a face of a planar slab waveguide, according to at least one embodiment of the present invention.

[0205] FIG. **13** is a schematic cross-sectional view of a face-lit waveguide illumination system portion, illustrating light injection into a thin planar waveguide, according to at least one embodiment of the present invention.

[0206] FIG. **14** is a schematic view of a face-lit waveguide illumination system portion, illustrating an exemplary method of attaching a light coupling optical element to a face of a planar waveguide, according to at least one embodiment of the present invention.

[0207] FIG. **15A** through FIG. **15F**, illustrate various exemplary configurations of a linear optical element, according to at least some embodiments of the present invention.

[0208] FIG. **16** illustrate an exemplary arrangement of optical elements and light sources with respect to a face of a window pane, according to at least one embodiment of the present invention.

[0209] FIG. **17** is a schematic perspective view and raytracing of a face-lit waveguide illumination system portion, showing a plurality of optical fibers attached to a face of a planar slab waveguide, according to at least one embodiment of the present invention. as well as faces **218** and **418** of FIG. **4** are not limited to having a linear generatrix in a cross-section and may be also be formed by any curved or segmented profile. The suitable profiles may be selected, for example, in response to the known angular distribution of light emitted by source **30** or to the transversal size of light emitting diode **32** or its collimating optics, if any. Alternatively or in addition to that, the surface profiles may be selected based on the thickness of pane **4** and configured for reducing the light spillage into the environment. Any of faces **18**, **218** or **418** may also be optionally mirrored to prevent light escape at lower-than-TIR incidence angles.

[0210] FIG. **5** illustrates an embodiment of system **2** in which linear optical element **6** is represented by a band or strip of optically transmissive prismatic film laminated onto surface **10** of window pane **4**. The prismatic film has a smooth surface facing surface **10** and a linear pattern of parallel microprisms facing outwardly from pane **4**. The microprismatic pattern may be formed in the surface of the film by any suitable method, such as, for example, embossing, microreplication, and the like. Each linear microprism may have two planar faces **84** and **88** so that the prism array is formed by a continuous sequence of alternating faces **84** and **88**.

[0211] Linear light source **30** comprising an array of LEDs **32** is tilted with respect to a normal to surface **10** in the illustrated cross-section so as to provide a skew incidence of the light beam into the light receiving face of window pane **4** and facilitate light injection at angles favorable for TIR propagation. Light source **30** is also provided with a cylindrical collimating lens **34** extending along the linear LED array. The transversal cross-sectional outline of lens **34** may be selected to provide a narrow-angle collimation for the divergent beam emitted by LEDs **32**. According to one embodiment, lens **34** may be selected from cylindrical lenses providing the Full Width Half Maximum (FWHM) angle of 30 degrees or less. According to one embodiment, FWHM may be less than 20 degrees and, more preferably, less than 10 degrees.

[0212] By way of example and not limitation, a continuous strip TIR optic with part numbers 10397 or 10398 commercially available from Carclo Optics may be employed for lens **34**. Such or similar continuous strip optic may be injection molded or extruded from PMMA (acrylics), polycarbonate, or any other suitable optically-clear plastic material.

[0213] According to an aspect of the present invention, referring to FIG. 5, the angles of each of the faces **84** and **88** with respect to the plane of surface **10** may be selected in a manner similar to selecting the angles of faces **14** and **18** of FIG. 2. More particularly, each face **84** should be configured to direct light into the medium of window pane **4** at such an incidence angle with respect to a normal to surfaces **10** and **12** which is greater than the TIR angle at said surfaces. Depending on the initial propagation angle of a light ray, such function may involve some refraction and bending the ray path accordingly.

[0214] It will be appreciated that, when the prismatic film forming linear optical element **6** is index matched to the refractive medium of pane **4**, light rays refracted by faces **84** will generally maintain about the same propagation angles upon entering the pane. On the other hand, some rays may be emitted by source **30** at sufficiently low out-of-plane angles which already make incidence angles with respect to a normal to surface **10** greater than the TIR angle. Therefore, such rays may be injected into pane **4** without appreciable bending in which case the light coupling function of optical element **6** may be limited to suppressing refraction at surface **10** and thus preserving the TIR incidence angles for the rays being injected into pane **4**.

[0215] Accordingly, in at least some embodiments, faces **84** do not have to be necessarily configured to refract the incident light and may be positioned about perpendicularly to the incident beam to simply admit light into the film material without appreciable refraction. By approximate matching the refractive index of the prismatic film material to that of pane **4** and by providing a good optical contact of the film with surface **10**, the refraction at surface **10** can be suppressed. In this case, light rays entering the prismatic about perpendicularly to faces **84** will be injected into pane **4** without change in propagation direction. Thus, when the initial angle of the incidence rays with respect to a normal to surface **10** is greater than the TIR angle characterizing said surface and the ray maintains its direction upon entering the medium of pane **4**, the condition of TIR at both parallel surfaces **10** and **12** will automatically be met and the injected light can further propagate in pane **4** in a waveguide mode.

[0216] Referring further to FIG. 5, the slope of each face **88** may be selected to minimize the chance of light injection at incidence angles lower than TIR with respect to a normal to surfaces **10** and **12**. In at least some configurations of the prismatic film, this may be achieved by minimizing the interaction of the face with the incident light beam emanated by source **30**. By way of example and not limitation, the apparent light receiving aperture of faces **88**, as viewed from source **30**, may be minimized by varying the slope of faces **88** with the distance from source and making each face **88** about parallel to the line of sight to source **30**. In one embodiment the slope of faces **8** with respect to surface **10** decreases with the distance from light source **30**. In one embodiment, the slope angle of farthestmost faces **88**, with respect to surface **10**, may be less than 10 degrees or so. Furthermore, it may be preferred that at least the farthestmost faces **88** have even lower slope angles with respect to surface **10**. This may have an added advantage of faces **88** acting as TIR reflectors

in secondary interactions with light rays which have been injected into pane **4** by the preceding portions of the prismatic film along the propagation path. It will be appreciated that each face **88** extending near parallel to surface **10** may still reflect a substantial part of light rays already propagating in pane **4** in a waveguide mode as opposed to the case where face **88** would have a relatively high slope angle with respect to surface **10**.

[0217] FIG. **6** illustrates an exemplary arrangement of linear light emitting and light injecting components of system **2** in an exploded perspective view. Window pane **4** of FIG. **6** exemplifies a framed glass window pane in a building facade or a glass pane of a storefront window.

[0218] Light-injecting optical element **6** is formed by a microstructured prismatic film similar to that of FIG. **5**. The prismatic film is laminated to a face window pane **4** in a bottom portion of the window. The orientation of the prismatic film is such that its longitudinal axis **44** is aligned parallel to the bottom edge of the window and the linear micro-prisms of the film also extend generally parallel to the bottom edge. The face-mounted prismatic film is firmly attached to the glass surface using a low-tack optically-clear adhesive so that there is provided a good optical contact between the film and pane **4**.

[0219] Light source **30** of FIG. **6** is formed by a strip of high-brightness LEDs **32** interconnected with each other in a linear LED array and mounted to a narrow heat-conducting printed circuit board (PCB) **90**. A metal channel **136** is attached to the back of PCB **90** with a good thermal contact. Channel **136** may have a dual function. For example, it may provide a structurally rigid encasement for the LED array and also act as a heat sink for the LEDs **32**.

[0220] Light source **30** is also associated with linear collimating lens **34** extending parallel to the longitudinal axis of the LED array. Lens **34** should also be properly positioned with respect to the LED strip so that it can intercept all or at least a substantial part of the divergent light beams emanated by individual LEDs and collimate said beams at least in a plane perpendicular to the longitudinal axis of the LED array.

[0221] Collimating lens **34**, light source **30** and channel **136** may be bundled together in a single linear light-emitting assembly. The light-emitting assembly may be further encased into a housing (not shown) with an opening for the emitted beam.

[0222] The light-emitting assembly of FIG. **6** is positioned slightly below the prismatic film attached to the face of pane **4**, parallel to the ground. Additionally, as shown in FIG. **6**, the light-emitting assembly is rotated around its longitudinal axis so that the beam emitted by source **30** and collimated by lens **34** is generally pointed from bottom up, towards the prismatic film and generally towards the opposing (top) edge of pane **4**, as indicated by direction **98**. The light emitting assembly or any of its components may be attached to the window's frame or directly to the face of pane **4**, for example, by using suction cups or adhesive. Other methods of fixing the light emitting components of system **2** in front of optical element **6** may include attaching to the building's structural elements adjacent to the window, such as walls, floor, ceiling, etc.

[0223] Depending on the configuration of the LED strip and on the thickness of pane **4**, the prismatic film may take the shape of a relatively narrow strip or tape in which the linear prismatic features extend along the longest film dimension. The width of the prismatic film may be selected, for example, based on Equation 3, and its length may approximate the length of the LED strip. However, it is noted that other parameters of systems **2** components may also be taken into account, such as the beam spread of source **30**, the size of individual LEDs **32**, the transversal size of the exit aperture of lens **34**, and the like.

[0224] The prismatic film is configured to receive the collimated light beam emanated by the light-emitting assembly and admit such light beam into the glass medium of pane **4** at angles permitting for TIR propagation within the pane. As explained in the above-discussed embodiments referring to FIG. **5**, the interaction of the collimated beam with light-injecting prismatic film may involve refraction and/or total internal reflection of rays by at least some prism facets and may further involve the suppression of refraction at surface **10**.

[0225] Depending on the optical clarity of the glass of pane **4**, light may propagate in a waveguide mode considerable distances along the vertical extent of the window panel. It will be appreciated that a fraction of the propagating beam may reach the vertical edges of pane **4** due to the beam divergence in the respective plane. However, it will also be appreciated by those skilled in the art that the respective light rays will generally have internal incidence angles onto the surfaces of the vertical edges greater than the TIR angle at these surfaces, as a result of beam refraction by the light-receiving facets of the prismatic film and as a result of pointing the light beam towards the top edge of the pane.

[0226] Therefore, unless the vertical edges of pane **4** are excessively rough or specially sanded, which may cause significant light scattering and decoupling from the waveguide mode, most of light rays reaching an edge of the glass pane will undergo TIR from its surface and will therefore remain in the pane. Accordingly, the vertical edges of the glass panel may participate in the waveguiding function of pane **4** and may further promote beam mixing, thus resulting a more homogeneous distribution of light within the pane. It will further be appreciated that, in the case illustrated in FIG. **6**, the prevailing direction of light propagation **98** in window pane **4** will generally point from the bottom edge to the top edge of the pane even when the beam divergence and reflections from vertical edges of the pane are considered.

[0227] Provided that pane **4** has a sufficient optical clarity and low light absorption at least in the desired wavelengths, light extracting features **20** (not shown in FIG. **6**) may be positioned along direction **98** at a considerable distance from the area where light is injected into the pane. Since mixing divergent light beams from a series of LEDs **32** in pane **4** effectively distributes light through the volume of pane's body and since multiple reflections from opposing surfaces **10** and **12** homogenize such light, a uniform illumination of light extracting features **20** may be achieved even when such features are distributed over a relatively large area of the pane. Accordingly, decoupling light from pane **4** using evenly-lit light extracting features **20** may produce a relatively uniform glow from the respective portions of the pane. Considering that light extracting features **20** will deplete light from pane **4** along direction **98**, such features may be configured to have a variable extraction rate depending on the distance from source **30**. For instance, when light scattering elements are used for light extraction in features **20**, the density of such scattering elements may be made increasing along direction **98**.

[0228] Referring to both to FIG. **5** and FIG. **6**, it is noted that this invention is not limited to employing microprismatic films for forming extremely low-profile optical elements **6**. At least in some embodiments, the prismatic film of FIG. **5** and FIG. **6** may be replaced, for example, by a holographic film having suitable light-turning and/or refraction suppressing functions and being capable of injecting at least a portion of light entering its surface into pane **4** at angles favorable for TIR. In further examples and embodiments, optical element **6** may also include diffracting films, light scattering films or any other kind of light-turning films which can be configured for light injection into a waveguide mode.

[0229] In one embodiment, optical element **6** may include a light transmitting film or a thin plate structure which has two or more layers and is configured to redirect light internally by embedded microstructure rather than externally by surface microstructures such as those illustrated in FIG. **5**. In a non-limiting example, the embedded microstructures may be formed by casting a liquid transparent polymer having a first refractive index over a microstructured surface of another polymer having a different second refractive index. It will be appreciated that such an overmolding or casting process will create a corrugated boundary between the different-index layers which replicates the shape of the surface microstructures of the second polymer. When the overmolded surface microstructures are represented by a parallel array of linear microprisms, the internal boundary will comprise an array of alternating facets, each facet representing a refractive or TIR-reflective optical interface that can bend light that enters such layered optical structure.

[0230] It is noted that such embedded microstructures may be formed by any suitable process that

forms two or more layers of different optically-clear materials with a non-planar, preferably corrugated boundary. Such microstructures may also be made using coextrusion of two or more films having different refractive indices. Various configurations of film-thickness optical structures that suitable for optical element **6** may include the light redirecting films disclosed in co-pending U.S. patent application Ser. No. 13/662,311, incorporated herein by reference in its entirety. [0231] FIG. **7** illustrates an embodiment of face-lit waveguide system **2** in which optical element **6** is exemplified by a two-layer optically transmissive plate having internal corrugated boundary with a prismatic (sawtooth) cross-sectional profile. It is noted, however, that the same type of corrugated boundary and essentially the same basic functionality may also be achieved using a film-thickness structure.

[0232] The plate of FIG. **7** is generally planar, has two opposing parallel surfaces **76** and **16** and is formed by a first optically transmissive layer **62** and a second optically transmissive layer **64**. The plate is configured for a generally unimpeded light passage through its body where surface **76** is configured as a light input surface and surface **16** is configured a light output surface.

[0233] Surface **16** is made optically smooth and attachable to the planar surface **12** of window pane **4** with a good optical contact. Layer **28** of index matched adhesive or encapsulant is provided to promote such optical contact and adhesion.

[0234] Similarly to FIG. **6**, light source **30** of FIG. **7** is exemplified by a linear array or strip of compact high-brightness LEDs **32**. The longitudinal axis of the linear array extends parallel to the longitudinal axis of prismatic corrugations of the internal layer-to-layer boundary of the plate. In a further similarity to FIG. **5**, source **30** also comprises collimating TIR lens **34**. Lens **34** has a linear extruded configuration and is designed to collect the divergent light beam emanated by LEDs **32** and collimate such beam to a narrower angle, directing it onto light input surface **76**.

[0235] As illustrated in FIG. **7**, source **30** and collimating lens **34** are positioned to illuminate surface **76** at an angle. The transversal aperture of lens **34**, its collimation angle and the tilt angle should be such that source **30** can illuminate substantially the entire light input surface **76** of the plate. Additionally, the direction of light emission should be generally oriented towards the intended prevailing direction **98** of light guiding in pane **4** and towards light extracting features **20** (not shown).

[0236] Referring further to FIG. **7**, layer **62** of optical element **6** has a first refractive index $n_{\text{sub.1}}$ and layer **62** has a second refractive index $n_{\text{sub.2}}$ which is higher than $n_{\text{sub.1}}$. It is preferred that there is at least a minimum relative difference in refractive indices between layers **62** and **64** to enable effective light bending. The refractive indices $n_{\text{sub.2}}$ and $n_{\text{sub.1}}$ as well as the angles of the prismatic facets of the boundary between the two layers should be configured to enable injecting light ray emanated by source **30** into window pane **4** at a relatively high angle with respect to a surface normal. Such angle should be greater than the TIR angle characterizing surface **10** and **12** and should provide for the subsequent light propagation in a waveguide mode by bouncing from opposing surfaces **10** and **12**.

[0237] In a non-limiting example, the prismatic boundary facets which are facing source **30** may be configured to intercept at least a substantial portion of the emitted light and perform the main light bending function of element **2**. As further illustrated in FIG. **7**, such facets can be fairly steep, making the dihedral angles with respect to the plane of pane **4** in the range from 85 to 90 degrees. The so high dihedral angles may be advantageously selected to most effectively bend light rays within the body of element **6** towards the plane of window pane **4** so that the minimum incidence angles needed for TIR at surface **10** can be reached for at least a substantial part of the light beam.

[0238] In contrast, the boundary facets which are facing away from source **30** may be configured to make substantially lower dihedral angles with respect to such the plane of pane **4**. Such dihedral angles may be further selected so that the respective facets are aligned along the prevailing path of light propagation in the body of element **6**. This may be useful, for example, for minimizing or eliminating the interaction of the non-functional boundary facets with light and particularly for

minimizing unwanted refractions and reflections from such facets which may result in light spillage.

[0239] It is noted that the internal boundary between layers **62** and **64** was shown in FIG. **7** with just several prismatic features for the illustrative purpose only and the illustrated representation of the boundary and the dimensions of optical element **6** should not be viewed as limiting this invention in any way. It should be understood that such boundary may also be formed by a large array of micro-scale prismatic facets which can be as small as a few micrometers across. Accordingly, the thickness of optical element **6** of FIG. **7** may also be varied in a broad range, from relatively thin and flexible films in the few-micrometer range up to rigid plates or slabs of several millimeters thick.

[0240] FIG. **8** illustrates an embodiment of face-lit waveguide system **2** in which linear collimating element **6** has curvilinear surface profiles in a transversal cross-section and where it also combines the light-injection and light-collimation functions. In the illustrated transversal cross-section, optical element **6** has a portion facing light source **30** where it is shaped as a TIR collimating lens. Such optical element **6** further features an opposing portion having planar face **16** attachable to the planar surface **10** of pane **4**. The curvilinear side walls **106** and **118** of element **6** should preferably have smooth polished surfaces providing for lossless TIR and should be configured to aid the light-collimating and light-injection functions. Either one or both walls **106** and **118** may also have planar portions or can be formed by any suitable conjugate of curved and planar profiles.

[0241] The extent of planar face **16** along the prevailing propagation path of waveguide light should generally not exceed a predetermined length which is defined by the maximum out-of-plane angle of the injected beam and by the thickness of pane **4**. Such length may be defined, for example by Equation 3 described above. Particularly, in an exemplary case of window pane **4** made from glass, the extent of planar face **16** along direction **98** should not exceed $1.8d$, where d is the glass pane thickness.

[0242] System **2** may incorporate any masking elements to block light which may escape from optical element **6** or from adjacent portions of pane **4** due to imperfect coupling. Some light may escape may occur, for example, due to some rays striking surfaces **10** or **12** at incidence angles lower than the critical angle of TIR. When applied to panel-type illuminated signage, masking the respective areas where the stray light may emerge from the system can provide an improved appearance and contrast of the images or patterns visualized by features **20**.

[0243] FIG. **9** depicts an embodiment of system **2** which includes a portion of window pane **4**, light source **30**, light-coupling optical element **6**, broad-area light extracting feature **20**, a reflector **42**, and light masking element **142**.

[0244] Pane **4** of FIG. **9** is made from highly transparent glass material and has sufficiently smooth surfaces **10** and **12** to provide for lossless TIR propagation along at least direction **98**. Such pane **4** may be exemplified by a portion of glass storefront window in which surface **10** is facing the interior of the store and opposing surface **12** is facing the outside.

[0245] Light source **30** includes a linear array of electrically interconnected LEDs **32**, structural channel **136** and heat sink **36** attached to channel **136**. Optical element **6** is exemplified by a strip of light transmitting film which is laminated onto surface **10** and configured to at least partially suppress the natural refraction at the air/glass interface. The light-coupling film may also be configured to bend at least rays that enter surface **10** at sub-TIR angles and communicate greater-than-TIR angles to such rays.

[0246] Reflector **42** is positioned generally parallel to the prevailing plane of pane **4** and is configured to reflect any uncoupled light rays that may emerge from pane **4** along the extent of the light coupling film. Such light may emerge from pane **4**, for example, as a result of decoupling of certain light rays by element **6** due to the secondary interactions with the refraction-suppressing film. The decoupled rays may be recycled by reflecting from reflector **42** and entering pane **4** again which may further enhance light coupling efficiency. Thus, reflector **42** positioned at surface **10**,

which is opposite surface **12** of pane **4**, may act as a light guiding channel aiding the light coupling process.

[0247] Light masking element **142** is exemplified by an opaque film or thin sheet positioned at the opposing face of pane **4**. Such masking element may be useful, for example, for blocking stray light and preventing the glare that may result from some light leakage in the light coupling area. Element **142** may be configured to absorb, back-scatter or reflect light emerging from surface **12**.

[0248] The broad-area light extracting feature **20** is attached to surface **10** of pane **4**. It is made from an opaque or translucent back-scattering material and is configured to extract light from pane **4** and direct the extracted light out of the pane through the opposing surface **12**.

[0249] In operation, window pane **4** acts as a face-lit waveguide delivery component for light which is emitted by source **30** and injected into the pane by optical element **6**. Pane **4** mixes and distributes the injected light so that at least a portion of it can be intercepted and extracted by light extracting feature **20**.

[0250] By using the above example of pane **4** being a storefront window with surface **10** facing inward, the extracted light can be emitted away from the store so that the glow of feature **20** may be viewed by an outside observer **660**. The inside location of feature **20** may have advantages of offering protection from the outside environment or elements. Additionally, the inside location of most or all light emitting and light management elements of system **2** may also have an added advantage of providing more convenience for the store personnel, easier serviceability and better protection from theft or vandalism.

[0251] Referring further to FIG. **9**, system **2** may be implemented in an alternative configuration where light extracting feature **20** is attached to the outside face or surface of pane **4** and configured to have forward-scattering properties. A yet further alternative may include configuring light extracting feature **20** to provide the visibility of its glow from both sides of pane **4**.

[0252] FIG. **10** depicts an embodiment of system **2** which includes a framed portion of glass window pane **4**, light source **30**, light-coupling optical element **6**, and broad-area light extracting feature **20**. Window pane **4** has two opposing broad-area faces **100** and **120** which are generally smooth and planar. An edge **700** of window pane **4** is covered by a frame member **360** and is not easily accessible for light input. Therefore, in the illustrated case, the broad area faces of window pane **4** represents a convenient alternative light input surface for injecting light into pane **4**. In the illustrated case, face **100** is chosen for light input.

[0253] Optical element **6** is formed by a rectangular planar plate or strip of light transmitting material. Direction **98** indicates the intended prevailing propagation path of light in window pane **4**. Optical element **6** is attached to face **100** with a good optical contact and aligned with respect to pane **4** so that the longitudinal axis of the rectangular planar strip extends perpendicular to direction **98** and parallel to face **100**.

[0254] Light source **30** included a plurality of LEDs **32** arranged in a linear array extending parallel to the longitudinal axis of optical element **6**. Each LED **32** is positioned in an immediate proximity to the light input face **14** of element **6** and also in a close proximity to face **100** of pane **4**. The optical axis of each LED **32** is aligned about parallel to the plane of face **100** so that the light input face **14** of element **6** is illuminated from a perpendicular direction.

[0255] Light extracting element **20** is attached to face **100** in a different location of pane **4** along direction **98**. Such element **20** may include retro-scattering film which carries an image print and is configured to emit light within a prescribed angular cone towards a perpendicular direction when illuminated from the inside of pane **4**. The light-scattering film of element **20** may be laminated onto face **100** with sufficient optical contact for providing light interaction with the respective light scattering features of the film.

[0256] In one embodiment, in order to maximize the efficiency of light input into pane **4** through its face **100**, the thickness of element **6** and the transversal size of source **30** may be selected to be considerably less than the thickness of the pane. In one embodiment, the transversal width of the

contact area of optical element **6** with face **100** may be selected from Equation 3.

[0257] In operation, referring further to FIG. **10**, the light beam emitted by source **30** enters light coupling optical element **6** through face **14** and further propagates through the refractive material of element **6**. It will be appreciated by those skilled in the art that, although each LED **32** may emit a highly divergent beam, the divergence angle will generally be narrower inside element **6** than in the air due to the difference in refractive indices. The refractive index of glass and many optically transmissive plastics is around 1.5 or greater while the refractive index of air is around 1. At such difference in refractive indices, practically all rays in of the light beam striking face **14** will bend toward the prevailing plane of element **16** sufficiently to enable further propagation along direction **98** by means of TIR.

[0258] As illustrated in FIG. **10**, some light rays entering element **6** through face **14** can propagate directly into pane **4** via the coupling face **16**. The optical contact between face **16** and face **100** suppresses refraction towards a surface normal which would normally occur at the air/glass interface of the bare surface of pane **4**. As a result, such light rays may enter pane **4** without appreciable changing the propagation direction and can thus maintain the greater-than-TIR incidence angle with respect to face **120**. Once reflected from face **120**, such rays will also reflect from the free portions of face **100** due to the parallelism of faces **100** and **120**.

[0259] As further illustrated in FIG. **10**, some light rays entering element **6** through face **14** will initially strike face **22** which is opposing face **16**. Such rays will reflect from face **22** by means of TIR and will subsequently enter window pane **4** via the coupling area formed by the optical contact of face **16** and face **100**. Likewise, due to the suppressed refraction at the light coupling area, such light rays will continue propagating within pane **4** in a waveguide mode.

[0260] Light rays propagating in a waveguide mode through window pane **4** may eventually strike the optical interface formed by light extracting element **20** and face **100**. Light extracting element **20** decouples such rays from pane **4** and directs them towards viewer's eye **660**. In one embodiment, light extracting element **20** may be configured to provide maximum intensity of the extracted light along a surface normal **220**. Depending on the application, element **20** may also be configured to emit light from pane **4** in a broad angle of angles with respect to normal **220** or emit a collimated beam towards the viewer. Light extracting element **20** may also be configured to emit light into opposing directions thus providing the viewability of its illuminated areas from both sides of pane **4**.

[0261] FIG. **11** shows a perspective schematic view of the embodiment illustrated in FIG. **10**. Light source **30** is exemplified by a side emitting flexible LED strip having four individual LEDS **32**. Each LED **32** each exemplified by a surface-mount LED device (SMD) which can be implemented in a fairly small package being less than 2 mm in height. The small size of SMD LEDS makes them particularly suitable for coupling to face **14** of element **6**. Such side-emitting flexible LED strips are available from various suppliers and may be based on a number of densely or sparsely packed SMD LEDs. For example, one type of side-emitting (also called side-view) LED strips is available from Elemental LED of Emeryville, CA (Stock number EL-IMGSDRIB12V). The side-emitting LED strip may be provided with an adhesive backing which can be used for attaching the strip to face **100** of pane **4**.

[0262] The light-transmitting rectangular plate or strip plate exemplifying optical element **6** may be attached directly to face **100** using an optical adhesive, optically clear double-sides adhesive tape or curable encapsulant. Alternatively, an intermediate light coupling film or plate may be used as a layer between the plate and pane **4**. In a further alternative, optical element **6** may be pressed against face **100** using an external fixture. The external fixture may be attached to frame **300** or to surface **100** of pane **4**. A suction device such as a pair of suction cups may also be used for attaching element **6** to face **100**. When element **6** is mounted to face **100** with some lasting pressure applied to it, a liquid or gel-type optical encapsulant may be used instead of a fully cured one, in which case element **6** may allow for its easier removability.

[0263] Since forming illumination system **2** does not generally require surface penetration into the waveguide plate and either one or both of optical element **6** and light extracting feature **20** may be implemented in the form of an easily attachable and detachable films or decals, this may provide additional utility of system **2** and lower its cost of ownership. Particularly, system **2** may be made easily deployable onto the existing windows inside or outside of the residential, commercial or governmental buildings without the need of undesired changes or invasion to the existing structures. Since at least some types of commercially available films used for in-window signage are easily removable from smooth surfaces, the film representing light extracting feature **20** may be made interchangeable without the need to replace the entire system. Thus, the illuminated pattern, its shape, color, size, and the like may be changed with a relative ease and at low cost by merely removing the light extracting film and laminating another one onto the same or any other area of pane **4**.

[0264] FIG. **12** schematically illustrates an embodiment of system **2** which employs a plurality of elongated optical elements **50** and where each element **50** has alternative configuration and orientation with respect to direction **98** compared to element **6** illustrated in the preceding Figures.

[0265] Referring to FIG. **12**, there is provided a planar waveguide **190** defined by opposing broad-area surfaces **410** and **412** and made from a highly transmissive glass or plastic material. Light source **30** of FIG. **12** is exemplified by a side-emitting flexible LED strip similar to that of FIG. **11**. Each of LEDs **32** is provided with an individual light coupling optical element **50** which has a form of a linear rod or bar which has a rectangular transversal cross-section and which longitudinal axis **44** extends parallel to the intended prevailing propagation direction **98** of light in pane **4**.

[0266] Each rectangular bar is made from an optically transmissive dielectric material and has four longitudinal sides each preferably having an optically smooth, polished surface. At least one terminal end of the rectangular bar also preferably has a smooth surface **92** configured for light input. One longitudinal side of each element **50** is disposed in optical contact with surface **410** of waveguide **190** and is configured for light output while the other three longitudinal sides are exposed to air and configured for reflecting light by means of TIR.

[0267] Such configuration and orientation of elements **50** allows each such element to receive light by its surface **92** and guide such light along its longitudinal axis **44** by reflections from the three exposed sides in response to TIR and optical transmission. Each element **50** therefore acts a light guide or light pipe for at least some rays which are not initially propagating towards the light output side. The light output side of element **50** provides an outlet through which light can leak from element **50** into waveguide **190** along the propagation path.

[0268] An advantage of implementing optical element **50** in the form of a light pipe may be that such configuration may allow for lower-loss light input into a thinner waveguide compared to, for example, the configuration illustrated in FIG. **11**. The use of arrangement of optical element **6** and LED-based source **30** of FIG. **11** for injecting light into a waveguide which have a thickness considerable less than the thickness of the plate representing element **6** may be problematic. This problem arises from the fact that a sizeable fraction of light may escape from element **6** through its face which is opposing the light input face **14**. In contrast, the optical element **50** of FIG. **11** may be configured to inject most light into the underlying waveguide even when the thickness of the waveguide is less than that of the respective rectangular rod. Since each LED **32** emits a light beam which has some divergence in the plane of waveguide **190**, at least a portion of said beam will escape from the TIR envelope of element **50** at each passage of the beam through the light output side of element **50**. The longer the longitudinal extent of the rectangular rod forming element **50**, the more light will be injected into waveguide **190** and the less light will remain within the TIR envelope.

[0269] The operation of light-guiding optical element **50** is further illustrated in FIG. **13** by example of light injection into waveguide **190** which thickness is less than the transversal dimensions of the rectangular light transmitting bar forming element **50**.

[0270] Referring to FIG. 13, light coupling optical element 50 is shown in a transversal cross-section. A face 94 of element 50 represents the side of the respective rectangular bar which is disposed in the optical contact with surface 410 of waveguide 190. A face 96 of element 50 represents an opposing side of the bar. Faces 91 and 93 respectively represent the other two sides of the bar in the transversal cross-section.

[0271] A light ray 336 exemplifies light received by element 50 from an individual LED 32 of light source 30. It will be appreciated that the optical contact of face 94 with surface 410 provides for a generally unimpeded light passage from optical element 50 into waveguide 190 and back. At the same time, it will be appreciated that the other three sides of element 50 as well as the portion of surface 412 of the waveguide immediately below element 50 form a TIR envelope (indicated by a dashed box area 222 in FIG. 13). Such TIR envelope confines ray 336 so that such ray propagates along the longitudinal axis of element 50 undergoing TIR from said faces and from surface 412. Ray 336 continues propagating through the body of element 50 until it reached an opening between face 93 and surface 412. Ray 336 thus eventually escapes from envelope 222 and continues its propagation in the body of waveguide 190 by means of TIR, which completes the injection of such ray into the waveguide.

[0272] It will be appreciated that, when all rays emitted by the respective LED 32 are considered, the prevailing direction of light propagation will be generally parallel to the longitudinal axis of the rectangular rod forming optical element 50. While some rays may reach the opposite end of the rod and escape from system 2 without being injected into waveguide 190, the percentage of such rays with respect to the entire emitted beam can be made relatively small.

[0273] There are several factors determining the amount of light which may remain in element 6 uncoupled into waveguide 190. One such factor is the relation of the thickness of waveguide 190 to the cross-sectional dimensions of optical element 50 as it determines the relative size of the gap in envelope 222 available for rays to escape. Another factor is the relative length of element 50 with respect to the waveguide thickness and with respect to the transversal dimensions of the optical element. Generally, the longer the linear element 50 is the higher percentage of light may escape from it. A yet another factor determining the escape rate from optical element 50 is the initial divergence of light beam emitted by the respective LED 32. A greater divergence angle, especially in the plane of waveguide 190, will generally result in a higher rate of light escape from optical element 50.

[0274] Referring to FIG. 12, the beam divergence should also be accounted for when selecting the spacing between adjacent pairs of LEDs 32 and optical elements 50. Particularly, it may be preferred that such spacing be sufficient to generally prevent or minimize interference between the adjacent LED-coupler pairs. In other words, light injected into waveguide 190 through one optical element 50 should not enter into another optical element 50 since it may cause ray decoupling and loss of illumination efficiency.

[0275] FIG. 14 provides a further illustration of the operation of rod-shaped optical element 50. A cone 710 denotes the divergence of the light beam in the plane of surface 410 as it escapes from element 50 into the underlying waveguide. LED 32 illuminating light input face 92 of element 50 emits light generally towards direction 98. Obviously, even when such light further undergoes multiple reflections within optical element 50 and enters into waveguide mode, the general propagation direction will be preserved. Thus, light injected into waveguide 190 using the above-described principles will continue propagating through the body of waveguide 190 along direction 98 and towards the predetermined location of the waveguide.

[0276] FIG. 14 also illustrates an exemplary method of attaching optical element 50 to surface 410 of waveguide 190. Accordingly, a rectangular piece of light transmitting film 780 is laminated onto surface 410 with a good optical contact. In a non-limiting example, the lamination may be done using a low-tack adhesive or static cling properties of the film. Optical element 6 may be attached to film 780 using a relatively high-tack adhesive. The adhesive may be applied in a liquid form or

in the form of a double-sided adhesive tape which should have sufficient optical clarity.

[0277] In case of using a low-tack adhesive of static cling for attaching film **780** to surface **410**, the film may generally have a lower peel resistance than optical element **50**. Therefore, in one embodiment, the area of film **780** can be made substantially larger than the area of the respective face **94** of optical element **50** which may provide a more reliable attachment of the film to the face of waveguide **190**.

[0278] FIG. **15A** through FIG. **15F** illustrate various exemplary configurations of linear optical elements **50** of FIG. **12**. Particularly, FIG. **15A** shows optical element shaped in the form of a rod having a square cross-section. In FIG. **15B**, optical element **50** has the shape of a rectangular bar. While light output face **94** of element **50** is shown to correspond to a longer side of the corresponding rectangle, it is noted that any of the shorter sides may be designated for this.

[0279] FIG. **15C** shows optical element **50** having the form of a half-round rod with one terminal end designated as light input face **92** and the planar longitudinal side designated as light output face **94**. Optical element **50** having a hexagonal rod configuration is shown in FIG. **15D**. In FIG. **15E**, optical element **50** is formed by a triangular optically transmissive. By way of examples and not limitations, the respective transversal cross-section may have the shape of a right-angle isosceles triangle or an equilateral triangle. However, it should be understood that such cross-section may be represented by any other type of triangle. In FIG. **15F**, optical element **50** is formed by a portion of a generally round rod where said rod has a planar longitudinal surface portion which is configured as light output face **94**.

[0280] It is noted, however, that possible variations of the transversal cross-sectional shapes of linear optical element **50** are not limited to the shapes illustrated in FIG. **15A** through FIG. **15F** or other preceding drawings. Optical element **50** may have an elongated configuration which transversal cross-section may approximate other common two-dimensional shapes including but not limited to pentagons, octagons, trapezoids, circles, circular segments or sectors, ovals, and the like.

[0281] FIG. **16** illustrates an embodiment of face-lit illumination system **2** which employs rod-shaped optical element **50**. The respective LED-based light sources **30** and elements **50** are grouped pairwise to create individual light emitting/coupling structures. Each pair is positioned at the respective corner of framed window pane **4** and is configured to inject light into the pane. Each optical element is positioned so that it directs light towards the respective direction **98** and within the respective angular cone **710** in the plane of pane **4**. All directions **98** converge towards the center of pane **4** so that the individual beams injected by source-coupler pairs are directed towards the central area of the window where light extracting feature **20** is located. Image-carrying light extracting area **20** is configured to extract light from pane **4** and direct the extracted light outwardly.

[0282] FIG. **17** depicts an embodiment in which light emitted by a single light source **30** is coupled into planar waveguide **190** through face **419** using a bundle **776** of optical fibers **60**.

[0283] Referring to FIG. **17**, light source **30** can be any light emitting device providing a beam of light that can be injected into a round aperture of the fiber bundle **776**. Particularly, it may be exemplified by a high brightness LED coupled to a collimating or concentrating optic which shapes the beam emitted by the LED according to the numeric aperture of the fibers **60** in bundle **776**. By way of example and not limitation, the LED optic may be represented by a 15-mm, hex shaped concentrator lens manufactured by Polymers Optics (Polymer Part #180) coupling into planar waveguide **190**. Such lens produces a 6 mm beam diameter at 14.5 mm distance in front of the lens and is designed particularly for inserting LED beams into fiber bundles.

[0284] Fiber bundle **776** includes five optical fibers **60**. In a non-limiting example, optical fibers **60** may be about 2 mm diameter each. The entrance aperture of bundle **776** is located at a prescribed location from the LED optic (e.g., approximately 14.5 mm in Polymer Optics concentrator lens is used) to intercept at least a substantial part of light emitted by source **30**.

[0285] Fibers **60** are distributed along an edge **806** of waveguide **190** and attached to surface **410** by means of light transmitting film **780**. Film **780** is laminated onto surface **410** with a good optical contact and sufficient adhesion to resist peeling off while holding fibers **60**. Fibers **60** are glued to film **780** in such a manner that there at least a portion of the surface of each fiber has sufficient optical contact with film **780** to enable light leakage into waveguide **190**. In one embodiment, the ends of each fiber **60** may be provided with a planar surface portion such as, for example, is shown in FIG. **15F**. In one embodiment, the end portion of each fiber may be flattened to facilitate bonding to film **780** and optical contact.

[0286] Each fiber **60** is straightened in the light coupling area and aligned perpendicularly to edge **806** so that the fiber ends attached to film **780** are generally parallel to each other and point towards the intended prevailing light propagation direction **98**.

[0287] The operation of the embodiment illustrated in FIG. **17** will be apparent in view of the above described embodiment of FIG. **12**. Accordingly, light emitted by source **30** and injected into fiber bundle **776** propagates in each fiber **60** in a waveguide mode without appreciable leakage until the area of film **780** is reached. The optical contact areas that each fiber **60** forms with film **780** result in light escape from the fibers and injection of such light into waveguide **190**. Moreover, it will be appreciated that, despite the divergence of individual light beams injected by each fiber **60**, the prevailing propagation direction **98** of the injected light in waveguide **190** will be generally parallel to the longitudinal axes of the straight ends of the fibers. Accordingly, suitable light extracting features may be positioned along direction **98** to extract light from a different location of waveguide **190**.

[0288] The foregoing embodiments have been described upon the case where the planar waveguide was exemplified by a highly transparent plate and a window pane which has a relatively high optical clarity. However, this invention is not limited to this and may be applied to the case where a relatively low-clarity or even translucent panel or plate may be used for distributing and emitting light injected through the face of such plate or panel. Many common objects may fall under this category, for example partially-transmitting glass or plastic elements of building interior or exterior, certain glass or plastic materials used for storefronts or building facades may have intrinsic light-scattering properties, etched door windows or glass table tops, etc. Alternatively, or in addition to that, various commercially available sheets of optically transmissive materials may have surface texture that scatters light propagating in the waveguide mode. Accordingly, although the distance of light propagation is shortened with reduced transparency, at least some of such plates, panels or sheets may act as face-lit planar waveguides extracting light along the entire propagation path and provide a soft glow from the entire surface or at least its portion adjacent to or surrounding the optical element **6**. Thus, lower-clarity planar optical waveguides may still be employed for system **2** in the manner proposed without the departure from the scope of this invention.

[0289] The structure and operating principles of the above described embodiments of face-lit illumination systems may be applied without limitations to any common glass or plastic objects which have the appropriate structure and sufficient optical clarity or transmissivity to act as planar waveguide. Examples include but are not limited to glass table tops, back-surface mirrors, glass or transparent-plastic doors or inserts of furniture articles, screens, light transmitting sheets employed in large-area lighting panels, light diffusing sheets, and the like. For instance, one or more optical elements **6** or **92** may be attached to a face of a glass table top or a vanity mirror. Each of the optical elements **6** or **92** may be provided with one or more light source **30** in order to insert light into the respective glass slab or panel and force its lateral propagation in a waveguide mode. A light extracting film may be attached to the face of such glass or panel in another location to extract light towards an observer. Either one or all of the attached optical components of system **2** may be made removable from the glass surface and may also be further made repositionable on the same or different surface. Additionally, different types of light coupling optical elements and light sources

may be used within the same system thus providing even greater degree of control over system configuration and light emission.

[0290] This invention is not limited in application to the planar light guiding panels, sheetforms or slabs having strictly parallel broad surfaces, but can also be applied to the case where the planar light guide has a wedge configuration which may be exemplified by pane **4** being slightly tapered towards one of the edges. A tapered configuration of the planar waveguide may be advantageously selected, for example, for an improved light extraction where light can be injected by elements **6** or **50** into the light guide along a direction pointing generally towards the tapered edge.

[0291] Furthermore, system **2** may incorporate any number of auxiliary layers serving various purposes, such as, for example, providing additional mechanical strength, environmental resistance, peel resistance, improved visual appearance, color, etc. Any optical interface between a layer formed by a lower refractive index transmissive medium and a layer formed by a higher refractive index transmissive medium may also be provided with an intermediate optically transmissive layer, for example, for promoting the optical contact or adhesion between the layers. The intermediate layer should preferably have a refractive index which is greater than the lower of the two refractive indices at the given optical interface.

[0292] System **2** may further incorporate various color filters, inks, dyes or other devices or substances that change the color of the extracted light. System **2** may also incorporate polarizing elements, fluorescent elements, fluorescent elements, light scattering or diffusing elements and the like, which may be provided as separate layers or incorporated into the bulk material of the optical elements of source **30**, the body of optical elements **6** or **50**, pane **4**, film **8** or its features **20**.

[0293] Further details of operation of waveguide illumination system **2** shown in the drawing figures as well as its possible variations will be apparent from the foregoing description of preferred embodiments. Although the description above contains many details, these should not be construed as limiting the scope of the invention but as merely providing illustrations of some of the presently preferred embodiments of this invention. Therefore, it will be appreciated that the scope of the present invention fully encompasses other embodiments which may become obvious to those skilled in the art, and that the scope of the present invention is accordingly to be limited by nothing other than the appended claims, in which reference to an element in the singular is not intended to mean “one and only one” unless explicitly so stated, but rather “one or more.” All structural, chemical, and functional equivalents to the elements of the above-described preferred embodiment that are known to those of ordinary skill in the art are expressly incorporated herein by reference and are intended to be encompassed by the present claims. Moreover, it is not necessary for a device or method to address each and every problem sought to be solved by the present invention, for it to be encompassed by the present claims. Furthermore, no element, component, or method step in the present disclosure is intended to be dedicated to the public regardless of whether the element, component, or method step is explicitly recited in the claims. No claim element herein is to be construed under the provisions of 35 U.S.C. 112, sixth paragraph, unless the element is expressly recited using the phrase “means for.”

Claims

1. A wide-area lighting emitting panel, comprising: an optically transmissive sheet having a first broad-area surface, an opposing second broad-area surface, a first edge, an opposing second edge parallel to the first edge, a third edge perpendicular to the first and second edges, a fourth edge parallel to the third edge, a light mixing region adjacent to the first edge and defining a pair of opposing smooth and planar surfaces extending parallel to one another and configured for guiding light using total internal reflection, and a two-dimensional light extraction region separated from the first edge by the light mixing region; a plurality of light extraction features distributed across the two-dimensional light extraction region with a density increasing along a direction from the

first edge to the second edge and configured for extracting light from the optically transmissive sheet towards opposite sides of the optically transmissive sheet; a reflector positioned generally parallel to the optically transmissive sheet and configured to reflect light being extracted from the optically transmissive sheet; a light diffuser; a flexible optically transmissive prism sheet positioned parallel to the optically transmissive sheet and comprising a first prism film having a first array of parallel linear prisms facing away from the optically transmissive sheet and longitudinally extending between two edges of the flexible optically transmissive prism sheet; a flexible LED strip located adjacent to one of the pair of opposing smooth and planar surfaces and extending parallel to the first edge, wherein the flexible LED strip comprises a flexible printed circuit having a major surface extending generally parallel to the first and second broad-area surfaces, and a linear array of electrically interconnected side-emitting LED packages mounted to the major surface such that a plane of a light emitting surface of each of the side-emitting LED packages is oriented perpendicular to the major surface; and a structural housing element at least partially encasing the linear array of electrically interconnected side-emitting LED packages and covering at least portions of each of the pair of opposing smooth and planar surfaces, the structural housing element comprising a linear channel formed from a metallic material and having an opening configured to provide a light passage from the linear array of electrically interconnected side-emitting LED packages towards the light mixing region; wherein at least one of the electrically interconnected side-emitting LED packages has a height of less than 2 mm, wherein at least some of the plurality of light extraction features are separated from one another and from each of the first, second, third, and fourth edges by smooth and planar surface portions of the optically transmissive sheet, and wherein at least one of the plurality of light extraction features comprises a surface relief microstructure.

2. A wide-area lighting emitting panel as recited in claim 1, wherein a width of the light mixing region measured in a direction perpendicular to the first edge is greater than a thickness of the optically transmissive sheet by at least 3.5 times.

3. A wide-area lighting emitting panel as recited in claim 1, wherein a back surface of the flexible LED strip is provided with an adhesive layer and is attached to the linear channel with a good thermal contact.

4. A wide-area lighting emitting panel as recited in claim 1, wherein a surface of the flexible LED strip is disposed in direct or indirect contact with one of the pair of opposing smooth and planar surfaces of the light mixing region.

5. A wide-area lighting emitting panel as recited in claim 1, wherein the flexible printed circuit of the flexible LED strip is bonded to one of the pair of opposing smooth and planar surfaces of the light mixing region using adhesive.

6. A wide-area lighting emitting panel as recited in claim 1, further comprising an opaque light control element positioned in a proximity of the light mixing region and configured to absorb, back-scatter or reflect light emerging from the light mixing region.

7. A wide-area lighting emitting panel as recited in claim 1, wherein at least one of the electrically interconnected side-emitting LED packages is located at an edge of the flexible printed circuit.

8. A wide-area lighting emitting panel as recited in claim 1, wherein the light mixing region comprises a linear array of optically transmissive light coupling elements formed in a surface of the optically transmissive sheet and distributed along the first edge according to a regular pattern.

9. A wide-area lighting emitting panel as recited in claim 1, wherein the light mixing region comprises a linear array of optically transmissive light coupling elements formed in a surface of the optically transmissive sheet and distributed along the first edge according to a regular pattern, and wherein at least one of the optically transmissive light coupling elements is formed by a rounded ridge oriented perpendicular to a longitudinal axis of the flexible printed circuit.

10. A wide-area lighting emitting panel as recited in claim 1, further comprising a fluorescent film material extending longitudinally and laterally along the optically transmissive sheet and being

approximately coextensive with the two-dimensional light extraction region, wherein the fluorescent film material comprises light scattering features and is configured to convert light from a first wavelength of visible spectrum to a second wavelength of visible spectrum, and wherein the second wavelength is longer than the first wavelength.

11. A wide-area lighting emitting panel as recited in claim 1, wherein the light mixing region comprises a highly elongated light coupling element formed from an optically transmissive dielectric material and longitudinally extending parallel to the flexible printed circuit.

12. A wide-area lighting emitting panel as recited in claim 1, wherein the flexible optically transmissive prism sheet comprises an adhesive layer, and a second prism film bonded to the first prism film using the adhesive layer and having a second array of parallel linear prisms facing the first prism film.

13. A wide-area lighting emitting panel as recited in claim 1, wherein at least one of the electrically interconnected side-emitting LED packages is configured to emit light with a variable intensity.

14. A wide-area lighting emitting panel as recited in claim 1, wherein at least a first one of the side-emitting LED packages is configured to emit light of a first color and a second one of the side-emitting LED packages is configured to emit light of a second, different color, each of the first and second side-emitting LED packages being individually controllable to vary its light intensity.

15. A wide-area lighting emitting panel as recited in claim 1, comprising a plurality of linear cylindrical lenses.

16. A wide-area lighting emitting panel as recited in claim 1, comprising a flexible optically transmissive prism film positioned parallel to the optically transmissive sheet.

17. A wide-area lighting emitting panel, comprising: an optically transmissive sheet having a first broad-area surface, an opposing second broad-area surface, a first edge, an opposing second edge parallel to the first edge, a third edge perpendicular to the first and second edges, a fourth edge parallel to the third edge, a light mixing region adjacent to the first edge, and a two-dimensional light extraction region separated from the first edge by the light mixing region; a plurality of light extraction features distributed across the two-dimensional light extraction region with a density increasing along a direction from the first edge to the second edge and configured for extracting light from the optically transmissive sheet towards opposite sides of the optically transmissive sheet; a reflector positioned generally parallel to the optically transmissive sheet and configured to reflect light being extracted from the optically transmissive sheet; a light diffuser; a flexible optically transmissive prism sheet positioned parallel to the optically transmissive sheet and comprising a first prism film having a first array of parallel linear prisms facing away from the optically transmissive sheet and longitudinally extending between two edges of the flexible optically transmissive prism sheet; a flexible LED strip located adjacent to a surface of the light mixing region and extending parallel to the first edge, wherein the flexible LED strip comprises a flexible printed circuit having a major surface extending generally parallel to the first and second broad-area surfaces and disposed in direct or indirect contact with the surface of the light mixing region, and a linear array of electrically interconnected side-emitting LED packages mounted to the major surface such that a plane of a light emitting surface of each of the side-emitting LED packages is oriented perpendicular to the major surface; and a structural housing element at least partially encasing the linear array of electrically interconnected side-emitting LED packages and covering at least a portion of the light mixing region, the structural housing element comprising a linear channel formed from a metallic material and having an opening configured to provide a light passage from the linear array of electrically interconnected side-emitting LED packages towards the light mixing region; wherein at least a portion of the light mixing region has smooth and planar surfaces which are parallel to one another and generally free from the light extraction features, wherein the light mixing region is configured for mixing and guiding light towards the light extraction region using optical transmission and total internal reflection from the smooth and planar surfaces, wherein at least one of the electrically interconnected side-emitting LED packages has a

height of less than 2 mm, wherein at least some of the plurality of light extraction features are separated from one another and from each of the first, second, third, and fourth edges by smooth and planar surface portions of the optically transmissive sheet, and wherein at least one of the plurality of light extraction features comprises a light-diffusing surface relief microstructure.
